Supporting Information for Target-specific diagnostics improve screening of river-temperature extremes, but transfer remains uncertain

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Text S1–S11; Figures S1–S4; generated Tables S1–S50. Full case-level and split-level results accompany the table-source CSV files; summary tables do not replace their denominators. Tables S25–S50 contain supplementary development, planning and source-qualification evidence; they do not constitute a new joint confirmation.

S1 Reference custody and exclusions

The opened-source reference and all old donor columns were inventoried before defining new eligibility. Identity normalization uses provider and original identifier; candidate aliases use coordinates and available polygon connections. Shared records connect network labels into overlap components. Old catchment polygons are incomplete, so European station disjointness is not universal hydrological disjointness. USGS-only transfer provides an additional river-flow-disjoint source restriction.

The HydroShare resource contains extremes, meteorology, flow and catchment attributes, not continuous daily water-temperature truth. Continuous Swiss and French values were acquired separately, after the random role assignment. Metadata and completeness can define eligibility but recovery error cannot. Ineligible components are excluded without redistributing their roles. Swiss source release states are filtered in the acquisition contract. French date and time fields are combined before local-day coverage is checked; qualification 4 is explicitly unqualified, not validated. Thus French values are project aggregates and are not promoted into the primary documented-mean scope.

For the original new-station experiment, the audited Austrian route was monthly-only and the Norwegian continuous route remained unresolved; neither

supplied continuous truth. The USGS catalogs screened in that recruitment did not add reference records to its cohort. The final new sample has 53 stations, with 21, 13 and 19 development/calibration/test stations. Prospective evaluation requires complete 2019–2022 target years and is smaller, especially in France. The separate backup-source investigation below retains its own eligibility and evidence status.

S1.1 Additional public backup-source qualification

A further metadata-led search considered 7 USGS natural-stream sites and 25 candidate temperature-channel pair-years. 0 pair-years passed all prespecified reference requirements. Qualification retained the original daily support (at least twenty distinct hours spanning twenty hours), three completely observed prior primary years, observed primary annual boundaries, complete backup coverage of primary gaps, and visible-sensor agreement (at least ninety overlap days; absolute mean difference at most 0.25 degrees and 95th percentile at most 0.5 degrees). Only approved unflagged native observations entered the main reference. Provisional values and access failures retain separate status. For Mississippi River at Clinton, 2019 meets the gap-coverage and history conditions but fails sensor agreement; its 208 primary daily deficits are therefore not promoted to a reference experiment. All candidate outcomes are retained under `public_backup_search/`.

Salt Creek legacy retrieval returned HTTP 503; the public replacement OGC API supplied 536,116 native observations on 16 pages, with approval and qualifier strings preserved. DST offsets are handled explicitly. Distinct series identifiers do not themselves verify separate sensor hardware or an identical sampling position. Exact station identities 01193500 and 05578100 already occur in the opened panel, and catchment/event independence for the remaining sites is unverified. The search is a reference-eligibility audit, not an independent recovery test.

A public CC-BY author archive also provides PRT metadata for twenty-four NEON stream sites. The inspected archive contains no digital surface-water thermistor subarchive, and its checked water-quality schema contains no measured temperature field. Only ZIP directories and bounded schema, location and issue metadata have been opened. The official monthly catalogue was not anonymously

accessible at acquisition (HTTP 403); underlying eligible pairs remain unknown. No costly information-acquisition action is inferred from these free-source searches.

S1.2 Prospective reference intake and conservative dependence screening

A subsequent public-reference intake is separate from every evaluated cohort above. BoM Water Data Online supplied 148 Australian station daily archives; the intake checked 869 native station-year intervals. 80 candidate station-years at 52 stations passed the fixed daily-quality and observed-sampling requirements. Their original polygon-based development/calibration/test component counts are 20/16/10. These are qualification counts, not independent confirmation units or method effects.

An additional metadata-only screen uses weak connectivity of every cached positive physical-node link, including shared upstream bifurcations and common downstream connections without a recruited gauge. Nonpositive outlet sentinels cannot merge distinct rivers. The 516 mapped candidate stations form 147 conservative hydrologic blocks. The mapped qualified references occupy 33 such blocks; 6 span original roles, and 1 qualified station remains pending topology review. Within-role counts become 16/15/9 and may share a block across roles, so they must not be summed as disjoint units. Conservative basin connectivity does not by itself prove shared-record leakage or independent regional weather. The source is the BoM Geofabric V3.3 network (`https://www.bom.gov.au/water/geofabric/`), attributed to the Commonwealth of Australia (Bureau of Meteorology), 2022. Original roles and quality qualifications are preserved; no new target prediction error has been opened.

For the future three-method error allocation, the ideal one-threshold zero-failure floor is 104 groups per calibration/evaluation role, hence at least 208 disjoint groups even before training and joint-power requirements. This is a necessary lower bound, not a recommended recruitment size. The mapped pool alone is insufficient under this conservative grouping; 984 unmapped candidates remain unresolved rather than counted as zero eligible. Prior-pool exclusions, regional-event dependence, verified process attributes and new-population joint planning are still required. Acquiring reference records has therefore not completed a new confirmation or an independently backed natural-gap experiment.

S2 Distinct experimental populations

Controlled recovery crosses feasible climatological overlap and length within the same reference year. Four realizations average placement noise; they do not multiply the number of independent histories. Warm controls restrict total missing warm-season days, while fragmentation holds their total budget but need not hold true support exposure fixed. Missing cells and fewer-than-three permitted training-year exclusions remain in job manifests.

The admission cohort uses short as well as long gaps and up to five empirical geometries, not the balanced long-gap mechanism cohort. The change was written before new calibration/test outcomes. The earlier controlled-classifier results remain archived diagnostic evidence; their larger gains are not used as primary admission findings.

Recorded-calendar replay transfers actual incomplete-year geometries to complete years. It is conditional on availability of qualifying incomplete calendars and complete counterpart years, not a census of all monitoring years. New qualifying replay calendars are French aggregates; no Swiss documented-mean natural-calendar estimate is available. Neither missing-day mechanisms nor temperatures originally hidden by the monitoring system are identified.

S3 Models, information and uncertainty

Matched quantile boosting uses 350 iterations, depth four, learning rate 0.06, minimum leaf size forty and quantile 0.9. Generic features contain gap geometry, season, visible donors, training climatological variability, historical record length, boundary observations, air availability and H30-estimator disagreement. Actual reconstruction OOF residual diagnostics, mask-matched historical H30 errors and direct H30 uncertainty are stronger decision-time alternatives, not hidden evaluation truth. Full window features add probabilistic support, its spread, peak-start spread/entropy, competition and interactions. Exact allowlists are in the analysis contracts; truth, true overlap and realized errors are forbidden predictors.

Spline models add a five-knot cubic basis to the same small information set. The two-point equivalence margin is distinct from five-point superiority. Conditional

source intervals lying inside the former do not demonstrate whole-procedure or cross-population equivalence.

Residual trajectories preserve observed target values. Training residuals are generated by holding out each permitted historical year as well as the current target year. Fourteen-day blocks cannot cross NaN boundaries. Short interpolation gaps may use complete blocks equal to their longest run only when no nominal block exists. Unsupported uncertainty is explicit missing information; it is not zero uncertainty. Ninety-percent scenario bands are not promised coverage and are evaluated as such.

Certificate references retain their declared minimum training support. Some locally fitted certificate heads are unavailable with the new development budget. Those are not successful zero-risk predictions and cannot establish an architectural win for another method. A selected certificate's delivered point and error differ from a certificate-derived ranking applied to the fixed priority point.

S4 Risk estimand and proof

For a fixed predictor and candidate threshold, let each deployment component contribute within-component mean admission A_g and failure-and-admission B_g. Since $0 \le B_g \le A_g \le 1$,

$$X_g = B_g + r(1 - A_g) \in [0, 1], \qquad r = 0.05.$$

The desired ratio is $E[B_g]/E[A_g]$, not mean component-conditional risk and not arbitrary case-weighted risk. For G independent deployment-representative components, invert the bounded-variable Chernoff–KL inequality to obtain an upper mean bound U satisfying

$$G\,\mathrm{kl}(\bar{X}, U) = \log(m/\alpha),$$

where m is the number of training-fixed candidate thresholds. A union bound covers candidate search, including nonmonotone risk. If any threshold has $U \le r$ and positive admission, then $E[B_g] \le rE[A_g]$ on the simultaneous-validity event. No result follows under arbitrary target shift or dependent components. The zero-mean upper bound is $1 - (\alpha/m)^{1/G}$, strictly positive.

Even the ideal all-admit, zero-failure observation therefore needs 59 groups for one candidate or 118 for twenty-one. Target calibration is too small. Empty certified admission is insufficient evidence, not observed risk zero or a recommendation to discard every record.

The empirical bootstrap rule is a separate descriptive operating policy. Constructed populations include twelve, forty and one hundred perfectly dependent-within-group cases and a rare high-risk group, with uninformative scores. These are counterexamples to general calibration claims, not measured frequencies in rivers. Test intervals describe variation over components conditional on fixed predictors; they cannot supply probability for an unobserved failure type.

S5 Freezing and time

Comparative source folds are 60/20/20 fitting/calibration/testing. Final source heads use an independent source calibration split, fitting on the other source components. F0 transfers the head and source threshold; F1 changes only target calibration; F2 trains on new development records and calibrates on the target calibration role. Provider-excluded and USGS-only restrictions remain distinct. Historical local reconstruction fitting is allowed by the stated deployment scenario.

The method lock predates target calibration errors, and the deployment lock predates test errors. No score, threshold, outcome margin or assigned role is changed after testing. The fully prospective global head/calibration use only years through 2018; test years are 2019–2022 and every upstream local input uses earlier years. Earlier-year local reconstruction with all-calendar source cross-fitting is labelled Past-local, not fully prospective.

Prospective in compact table labels denotes calendar-time forward hindcasting. Inputs are current-vintage finalized observations and reanalysis; their original publication/revision vintages were not archived. The experiment does not establish actual as-of-2018 data availability or real-time field deployment. Predictor and local training dates remain strictly earlier; no frozen prediction changes to make this qualification.

S6 Reconstruction references

The CSDI reference has two layers, two heads, 32 channels, fifty diffusion steps, a 366-step single feature, mixed random/structured masks, maximum thirty epochs and patience five. It is trained on documented old-source records only, with independent old validation groups and one metadata-chosen validation station-year per group. Target normalization excludes the evaluated year. Eight conditional draws provide a pointwise median series and a secondary median of trajectory H30 values. It is a fixed retrospective reference, not optimized state of the art or a prospective model.

Air2stream-3/8 use the existing published-equation implementation with compiled numerical kernels verified against it. Parameters fit the longest permitted contiguous forcing segment excluding the target year, and strictly before it prospectively. Discharge normalization uses training forcing only. Each interior gap starts from the last observed left anchor, never hidden truth. Incomplete forcing, unavailable discharge, insufficient training and nonfinite predictions are explicit statuses. Optimizer status and training error are retained. Comparisons show input requirements, availability and fixed-policy error on exactly the same supported cases.

S7 Decisions, unknowns and representation

All operating policies begin with an identical candidate pool within an assay. Unknown or unsupported answers remain in its denominator. Local hot thresholds use permitted training-year H30 quantiles; fixed 18 and 20-degree sensitivity thresholds are not regulatory claims. Full false-positive/negative, correct/unknown hot/cool counts, near-threshold outcomes and network coverage are in original-pool CSVs.

The cost grid charges every unknown record and assumes no perfect repair. Intervals are conditional descriptive component-bootstrap intervals, not simultaneous inference across all costs or measured ecological values. Selected hot-frequency error is distinct from original-pool unknown-label bounds. Those bounds allow all unknown labels to be cool or hot and do not correct mistakes in reported labels.

A post-test deterministic baseline audit adds always-unknown, always-hot and always-cool policies to the unchanged cost grid. Always unknown has loss exactly

the abstention cost; it produces no magnitude prediction and no finite selected-risk estimate. This does not alter the original H3 comparator, but prevents an improvement over a weak reconstruct-all reference being presented as operational superiority. Point dominance in selected sensitivity cells is not a new confirmatory endpoint.

The largest fully observed window mean is a deterministic lower bound on the complete recorded maximum. It can retain a hot label without accepting H30 magnitude, but cannot establish a cool label. An eighty-percent mean is not a bound. Fixed observation margins are sensitivity scenarios, not provider error specifications. The extension changes no magnitude error event.

S8 Power, stopping and a nonpredictive correction

Each protocol has 1,000 complete learner/calibration refits. Entire overlap components are resampled, preserving all station histories and masks. Deterministic local fits are cached because their inputs do not change under whole-component resampling; global heads and thresholds are refitted. Evaluation components are also resampled for whole-procedure intervals. Learning curves use twenty subsets per non-full size; full-pool seed repeats have one distinct subset.

Power projections compare predeclared five-, eight- and ten-point alternatives with zero and five-point null margins. They assume source-like group distributions and do not establish common-risk feasibility. An effect exactly at a superiority boundary has only size-level rejection, not high power. Training/calibration uncertainty limits projections even with more test groups. Recruitment did not depend on whether a preferred method won.

An initial descriptive label mistakenly treated snow percentages as fractions. Correcting majority snow to fifty percent, with glacier influence first and unknown attributes explicit, affects no feature, fitted model, calendar, eligibility, component or frozen role. Original locks and outputs remain unchanged. Script 373 supplies corrected descriptive subgroup tables; raw script-360 hydrological labels are superseded. A descriptive near-natural string in an earlier case manifest is not verified regulation information and supports no unregulated-catchment claim. The dated erratum records when this correction was made.

S9 Generated evidence tables

Tables are generated by script 369 from result rows. Full case/seed outputs, unsupported head statuses, exclusions and table-source CSVs are authoritative when a compact table suppresses detail. Dashes mean unavailable or nonestimable, never demonstrated zero risk.

Table S1: Opened-source admission cohort. Network labels are not independent when target or donor records overlap. Documented daily means define the primary measurement scope; unclear daily statistics and project-aggregated records are sensitivity analyses.

Provider	Daily statistic	Labels	Stations	Cases
arso	undocumented	12	47	4394
chmi	project_aggregated_from_subdaily	17	73	8849
foen	daily_mean	2	11	3131
gkd_bayern	daily_mean	9	53	13818
hubeau	project_aggregated_from_subdaily	4	9	801
lubw	daily_mean	7	20	2223
nve_hydapi	daily_mean	17	41	3013
rws_waterwebservices	project_aggregated_from_subdaily	2	5	359
syke	non_daily_reference_reading	2	3	771
usgs	daily_mean	121	319	32652

Table S2: New-to-project continuous reference records after all-old-target/donor and metadata-only catchment exclusions. Roles were randomly assigned before daily values were acquired and never reshuffled. French daily means are project aggregates and include explicitly unqualified source observations. Institutional novelty and station novelty are not the same.

Provider	Role	Stations	Components
foen	calibration	9	9
foen	development	11	11
foen	test	7	7
hubeau	calibration	4	4
hubeau	development	10	10
hubeau	test	12	12

Table S3: Primary documented-mean controlled cells. Four start realizations are attempted per feasible station-year cell; an impossible cell is not fabricated. Case counts do not measure independent replication.

Climate overlap	Days	Masks	Station-years
full	30	19536	4720
none	30	19536	4720
partial	30	19536	4720
full	60	19536	4720
none	60	19536	4720
partial	60	19536	4720
full	90	19536	4720
none	90	19536	4720
partial	90	19536	4720

Table S4: Paired full-minus-zero climatological overlap changes in absolute H30 error, averaging start realizations within a station-year before component bootstrap. Warm-season controls hold total missing warm-season days fixed. Intervals are 95 percent component-bootstrap intervals, not causal effects of an unobserved missingness mechanism.

Method	Control	Days	Change C	Lower	Upper	Groups
Daily climate	All seasons	30	0.622	0.584	0.656	144
Daily climate	All seasons	60	0.836	0.775	0.894	144
Daily climate	All seasons	90	0.896	0.824	0.964	144
Smooth climate	All seasons	30	0.628	0.589	0.669	144
Smooth climate	All seasons	60	0.860	0.800	0.920	144
Smooth climate	All seasons	90	0.926	0.849	0.996	144
Interpolation	All seasons	30	0.411	0.385	0.443	144
Interpolation	All seasons	60	1.032	0.980	1.090	144
Interpolation	All seasons	90	1.785	1.700	1.867	144
Neighbour ridge	All seasons	30	0.280	0.240	0.327	127
Neighbour ridge	All seasons	60	0.377	0.332	0.430	126
Neighbour ridge	All seasons	90	0.407	0.359	0.465	124
Air ridge	All seasons	30	0.516	0.449	0.593	113
Air ridge	All seasons	60	0.753	0.674	0.847	113
Air ridge	All seasons	90	0.817	0.733	0.914	113
Daily climate	Warm season	30	0.564	0.531	0.599	143
Daily climate	Warm season	60	0.678	0.637	0.719	143
Daily climate	Warm season	90	0.694	0.559	0.811	28
Smooth climate	Warm season	30	0.571	0.533	0.612	143
Smooth climate	Warm season	60	0.703	0.658	0.756	143

Method	Control	Days	Change C	Lower	Upper	Groups
Smooth climate	Warm season	90	0.695	0.558	0.807	28
Interpolation	Warm season	30	0.371	0.342	0.407	143
Interpolation	Warm season	60	0.904	0.847	0.971	143
Interpolation	Warm season	90	1.464	1.055	1.858	28
Neighbour ridge	Warm season	30	0.256	0.216	0.304	126
Neighbour ridge	Warm season	60	0.319	0.276	0.370	124
Neighbour ridge	Warm season	90	0.469	0.327	0.621	24
Air ridge	Warm season	30	0.458	0.392	0.538	113
Air ridge	Warm season	60	0.614	0.529	0.721	113
Air ridge	Warm season	90	0.796	0.587	0.995	20

Table S5: Paired 90-minus-30-day H30 error at high and fully hidden true support. The 0.999 cutoff denotes the exact all-hidden discrete case; the comparison remains conditional on feasible masks and eligible station-years. Length is not negligible for every reconstruction.

Method	Hidden cutoff	Change C	Lower	Upper	Station-years	Groups
Daily climate	0.900	0.147	0.121	0.172	2454	144
Daily climate	0.999	0.133	0.097	0.166	396	98
Smooth climate	0.900	0.159	0.132	0.185	2454	144

Method	Hidden cutoff	Change C	Lower	Upper	Station-years	Groups
Smooth climate	0.999	0.145	0.106	0.179	396	98
Interpolation	0.900	1.304	1.225	1.391	2454	144
Interpolation	0.999	1.235	1.120	1.347	396	98
Neighbour ridge	0.900	0.043	0.035	0.053	1685	120
Neighbour ridge	0.999	0.032	0.016	0.051	251	76
Air ridge	0.900	0.138	0.114	0.169	1889	113
Air ridge	0.999	0.123	0.093	0.156	313	76

Table S6: Fragmented-minus-contiguous contrasts with equal total missing warm-season days. True hidden support is not held fixed; its change is shown rather than silently attributing the entire effect to fragmentation. Policy eligibility uses longest run, not total missing days.

Method	Days	Error change C	Support change	Lower	Upper
Daily climate	30	-0.271	-0.254	-0.288	-0.252
Daily climate	60	-0.262	-0.273	-0.286	-0.238
Daily climate	90	-0.246	-0.260	-0.274	-0.218
Smooth climate	30	-0.274	-0.254	-0.293	-0.254
Smooth climate	60	-0.267	-0.273	-0.291	-0.240
Smooth climate	90	-0.253	-0.260	-0.279	-0.225
Interpolation	30	-0.242	-0.254	-0.257	-0.229
Interpolation	60	-0.597	-0.273	-0.638	-0.556
Interpolation	90	-1.395	-0.260	-1.485	-1.304
Neighbour ridge	30	-0.115	-0.253	-0.135	-0.097
Neighbour ridge	60	-0.105	-0.272	-0.122	-0.090
Neighbour ridge	90	-0.101	-0.251	-0.118	-0.088
Air ridge	30	-0.213	-0.248	-0.243	-0.187
Air ridge	60	-0.231	-0.271	-0.259	-0.205
Air ridge	90	-0.217	-0.256	-0.249	-0.191

Table S7: Exact decomposition and sign audit. A positive switch increment is often compensation for underestimation, not a fabricated hot peak. Numerical identity residuals below the displayed precision are not evidence of physical truth accuracy.

Method	Cases	Switches	Still low share	Max residual C
Daily climate	312825	188204	0.604	0.000
Smooth climate	312825	188373	0.632	0.000
Interpolation	312825	169266	0.710	0.000
Neighbour ridge	225103	86886	0.550	0.000
Air ridge	239231	125468	0.707	0.000

Table S8: Two-by-two residual intervention: original, permuted, mean removed, and both. The interaction is original minus permuted minus centered plus both. Truth is required for all these interventions. Their percentage improvements cannot be added as identified physical contributions. Deployable visible-anchor adjustment is evaluated separately in the text.

Method	Intervention	MAE / interaction C	Lower	Upper	Groups
Daily climate	raw	0.450	0.412	0.486	144
Daily climate	permuted	0.398	0.370	0.425	144
Daily climate	demeaned	0.215	0.193	0.236	144
Daily climate	demeaned_permuted	0.071	0.068	0.075	144
Daily climate	interaction	-0.091	-0.101	-0.079	144
Smooth climate	raw	0.459	0.420	0.496	144
Smooth climate	permuted	0.397	0.368	0.424	144
Smooth climate	demeaned	0.221	0.200	0.243	144
Smooth climate	demeaned_permuted	0.069	0.065	0.073	144

Method	Intervention	MAE / interaction C	Lower	Upper	Groups
Smooth climate	interaction	-0.091	-0.101	-0.079	144
Interpolation	raw	0.521	0.497	0.545	144
Interpolation	permuted	0.461	0.440	0.483	144
Interpolation	demeaned	0.223	0.211	0.235	144
Interpolation	demeaned_permuted	0.072	0.069	0.075	144
Interpolation	interaction	-0.092	-0.098	-0.085	144
Neighbour ridge	raw	0.186	0.164	0.215	128
Neighbour ridge	permuted	0.165	0.144	0.192	128
Neighbour ridge	demeaned	0.076	0.066	0.087	128
Neighbour ridge	demeaned_permuted	0.027	0.024	0.030	128
Neighbour ridge	interaction	-0.028	-0.034	-0.023	128
Air ridge	raw	0.388	0.350	0.433	113
Air ridge	permuted	0.318	0.292	0.345	113
Air ridge	demeaned	0.172	0.157	0.191	113
Air ridge	demeaned_permuted	0.047	0.043	0.051	113
Air ridge	interaction	-0.055	-0.062	-0.047	113

Table S9: Same-point source admission with disjoint 60/20/20 fitting, calibration and evaluation components. Entries are median [minimum, maximum] over three split seeds, not confidence intervals or independent experiments. Past-local recomputes local inputs from earlier years but does not freeze the global head in calendar time. All scores share the same empirical 5 percent calibration target, not identical realized test risk.

Score	Protocol	Retention percent	Selected failures percent
Certificate proxy	Retro	72.7 [72.6, 73.7]	2.6 [2.5, 2.9]
Complete diag.	Retro	78.8 [78.1, 79.0]	2.7 [2.6, 2.7]
Daily disagreement	Retro	1.0 [0.0, 1.8]	4.1 [3.3, 4.9]
Raw H30 uncertainty	Retro	77.8 [77.2, 79.3]	2.8 [2.6, 3.2]
Generic + OOF	Retro	78.1 [78.0, 79.1]	2.7 [2.6, 3.0]
Generic daily	Retro	71.2 [70.2, 72.4]	3.0 [2.8, 3.2]
Direct H30	Retro	78.9 [77.8, 79.6]	2.7 [2.5, 3.0]
Generic + overlap	Retro	76.4 [75.5, 77.7]	2.9 [2.6, 3.3]
Generic H30	Retro	76.6 [74.0, 77.4]	3.0 [2.4, 3.3]
Historical H30	Retro	72.8 [72.8, 73.6]	2.8 [2.8, 2.9]
Length	Retro	36.8 [36.7, 37.9]	1.9 [1.8, 1.9]
OOF daily scale	Retro	32.6 [24.3, 36.9]	2.4 [1.6, 3.0]
Climate overlap	Retro	67.6 [66.9, 68.7]	3.0 [2.9, 3.1]
Policy certificate	Retro	51.1 [50.7, 51.3]	2.4 [2.4, 2.6]
Mean support	Retro	69.7 [68.6, 69.9]	2.7 [2.4, 2.7]
Overlap x residual	Retro	75.5 [74.3, 75.7]	3.1 [2.7, 3.1]
Season spline	Retro	72.6 [71.9, 74.1]	2.9 [2.9, 3.3]
Spline + overlap	Retro	72.8 [71.6, 73.0]	2.9 [2.7, 3.0]
Small logit	Retro	64.2 [59.4, 66.6]	3.3 [2.8, 3.8]
Logit + overlap	Retro	69.9 [68.9, 69.9]	2.9 [2.9, 3.3]
H30 disagreement	Retro	70.8 [70.2, 71.5]	3.1 [2.8, 3.2]
Window	Retro	78.6 [77.8, 79.0]	2.6 [2.4, 2.8]
No competition	Retro	78.5 [77.7, 80.0]	2.6 [2.4, 3.1]
No donor features	Retro	79.5 [79.0, 79.8]	2.9 [2.8, 3.0]

Score	Protocol	Retention percent	Selected failures percent
No window spread	Retro	78.5 [77.8, 80.0]	2.6 [2.4, 3.1]
Upper support	Retro	69.9 [69.8, 70.0]	2.4 [2.4, 2.4]
Warm coverage	Retro	52.5 [49.7, 53.5]	2.2 [1.6, 2.4]
Certificate proxy	Past-local	69.4 [67.8, 69.7]	2.7 [2.5, 2.7]
Complete diag.	Past-local	76.0 [75.0, 77.5]	2.5 [2.4, 3.0]
Daily disagreement	Past-local	0.0 [0.0, 1.7]	7.8 [7.8, 7.8]
Raw H30 uncertainty	Past-local	74.4 [73.1, 74.8]	2.9 [2.6, 3.1]
Generic + OOF	Past-local	75.0 [74.0, 76.3]	2.5 [2.3, 2.9]
Generic daily	Past-local	66.0 [61.3, 66.2]	3.1 [2.5, 3.1]
Direct H30	Past-local	76.4 [75.7, 76.6]	2.7 [2.7, 2.9]
Generic + overlap	Past-local	72.0 [69.3, 73.5]	2.5 [1.9, 2.9]
Generic H30	Past-local	71.0 [69.0, 71.6]	2.2 [2.0, 2.6]
Historical H30	Past-local	62.9 [58.4, 64.1]	2.8 [2.1, 2.9]
Length	Past-local	37.4 [31.0, 37.5]	2.2 [1.8, 2.2]
OOF daily scale	Past-local	28.5 [21.7, 29.0]	2.3 [1.6, 3.0]
Climate overlap	Past-local	62.6 [59.5, 63.2]	3.0 [2.6, 3.4]
Policy certificate	Past-local	51.1 [48.4, 51.3]	2.4 [1.8, 2.5]
Mean support	Past-local	65.0 [62.6, 65.2]	2.7 [2.2, 2.9]
Overlap x residual	Past-local	69.9 [69.8, 70.1]	3.2 [3.0, 3.7]
Season spline	Past-local	70.1 [70.0, 70.7]	3.1 [3.0, 3.1]
Spline + overlap	Past-local	70.8 [70.1, 71.8]	3.2 [3.0, 3.5]
Small logit	Past-local	56.1 [55.9, 58.8]	3.0 [2.6, 3.5]
Logit + overlap	Past-local	62.8 [60.6, 63.6]	3.4 [2.7, 3.5]
H30 disagreement	Past-local	66.9 [65.9, 67.9]	2.7 [2.6, 2.9]
Window	Past-local	76.6 [75.1, 76.8]	2.7 [2.3, 2.7]
No competition	Past-local	76.6 [76.4, 76.7]	2.7 [2.7, 2.8]
No donor features	Past-local	77.2 [75.6, 77.7]	3.0 [2.4, 3.0]
No window spread	Past-local	76.4 [75.4, 76.6]	2.7 [2.4, 2.7]
Upper support	Past-local	65.5 [65.2, 67.4]	2.5 [2.4, 2.8]
Warm coverage	Past-local	48.1 [47.6, 48.3]	1.8 [1.7, 1.8]

Table S10: Conditional-on-fitted-procedure paired retention differences in percentage points. The primary superiority margin is 5 points and equivalence margin is 2 points. Crossing zero is not equivalence; a small conditional interval does not establish whole-procedure equivalence.

Comparison	Protocol	Seed	Difference	Lower	Upper
Window - Generic H30	Retro	101	1.698	0.247	3.034
Window - Direct H30	Retro	101	-0.501	-1.100	0.018
Window - Complete diag.	Retro	101	0.046	-0.669	0.722
Window - Generic + OOF	Retro	101	-0.026	-0.877	0.743
Generic H30 - Generic daily	Retro	101	6.176	5.100	7.361
Generic + overlap - Generic H30	Retro	101	0.350	-0.303	1.041
Logit + overlap - Small logit	Retro	101	9.541	6.882	12.776
Spline + overlap - Season spline	Retro	101	-1.247	-2.104	-0.486
Window - Generic H30	Retro	202	3.828	2.198	5.301
Window - Direct H30	Retro	202	-0.007	-0.271	0.261
Window - Complete diag.	Retro	202	-1.027	-1.850	-0.268
Window - Generic + OOF	Retro	202	-0.164	-0.771	0.492
Generic H30 - Generic daily	Retro	202	3.839	2.920	4.877
Generic + overlap - Generic H30	Retro	202	2.447	1.820	3.107
Logit + overlap - Small logit	Retro	202	5.662	3.275	8.309
Spline + overlap - Season spline	Retro	202	-1.012	-1.689	-0.394
Window - Generic H30	Retro	303	2.050	0.788	3.233

Comparison	Protocol	Seed	Difference	Lower	Upper
Window - Direct H30	Retro	303	-0.268	-0.676	0.139
Window - Complete diag.	Retro	303	0.527	-0.276	1.246
Window - Generic + OOF	Retro	303	0.467	-0.471	1.403
Generic H30 - Generic daily	Retro	303	4.132	2.723	5.716
Generic + overlap - Generic H30	Retro	303	-1.107	-1.658	-0.567
Logit + overlap - Small logit	Retro	303	3.366	1.543	5.522
Spline + overlap - Season spline	Retro	303	1.111	0.657	1.648
Window - Generic H30	Past-local	101	5.212	2.868	7.346
Window - Direct H30	Past-local	101	0.148	-0.435	0.885
Window - Complete diag.	Past-local	101	-0.768	-1.628	-0.093
Window - Generic + OOF	Past-local	101	0.472	-0.123	0.943
Generic H30 - Generic daily	Past-local	101	5.415	3.617	7.390
Generic + overlap - Generic H30	Past-local	101	1.900	1.041	2.827
Logit + overlap - Small logit	Past-local	101	4.689	3.245	6.226
Spline + overlap - Season spline	Past-local	101	1.725	0.611	3.131
Window - Generic H30	Past-local	202	7.622	5.526	9.306
Window - Direct H30	Past-local	202	0.241	-0.039	0.544
Window - Complete diag.	Past-local	202	1.566	0.513	2.601
Window - Generic + OOF	Past-local	202	2.624	1.297	4.385
Generic H30 - Generic daily	Past-local	202	7.721	6.608	9.039

Comparison	Protocol	Seed	Difference	Lower	Upper
Generic + overlap - Generic H30	Past-local	202	0.345	-0.116	0.775
Logit + overlap - Small logit	Past-local	202	7.477	6.148	9.026
Spline + overlap - Season spline	Past-local	202	0.825	0.270	1.451
Window - Generic H30	Past-local	303	4.149	1.464	6.589
Window - Direct H30	Past-local	303	-0.581	-1.442	0.338
Window - Complete diag.	Past-local	303	-0.861	-1.961	0.078
Window - Generic + OOF	Past-local	303	0.123	-0.628	0.839
Generic H30 - Generic daily	Past-local	303	4.971	3.224	6.699
Generic + overlap - Generic H30	Past-local	303	0.993	-0.502	2.481
Logit + overlap - Small logit	Past-local	303	4.009	2.538	5.589
Spline + overlap - Season spline	Past-local	303	-0.581	-1.213	-0.060

Table S11: Whole-procedure uncertainty from 1,000 network resamples, refitting four risk learners and recalibrating thresholds. The prospective global head and calibration use years through 2018 and evaluation uses 2019–2022. Cached local fits are unchanged under resampling their whole component.

Comparison	Protocol	Difference pp	Lower	Upper	Test groups
Window - Generic H30	Retro	2.595	-1.900	7.970	29
Direct H30 - Generic H30	Retro	2.535	-2.344	8.176	29
Window - Direct H30	Retro	0.059	-3.787	4.436	29
Direct H30 - Generic + OOF	Retro	1.800	-3.239	7.689	29

Comparison	Protocol	Difference pp	Lower	Upper	Test groups
Window - Generic H30	Prospective	2.916	-5.750	11.269	24
Direct H30 - Generic H30	Prospective	2.689	-6.952	10.955	24
Window - Direct H30	Prospective	0.226	-5.554	5.898	24
Direct H30 - Generic + OOF	Prospective	-1.466	-9.116	5.050	24

Table S12: Projected power to exceed the five-point practical margin under predeclared 8- and 10-point location-shift alternatives. Projections assume source-like independent groups and do not establish common-risk feasibility under transfer. Neither 80 nor 90 percent power is reached in this tested range; increasing only test groups leaves training/calibration variability. An effect exactly at the null margin cannot have high superiority power.

Protocol	Test groups	Assumed gain	Power
Retro	7	0.080	0.109
Retro	7	0.100	0.240
Retro	19	0.080	0.193
Retro	19	0.100	0.362
Retro	53	0.080	0.254
Retro	53	0.100	0.417
Retro	100	0.080	0.274
Retro	100	0.100	0.399
Retro	400	0.080	0.291
Retro	400	0.100	0.410
Prospective	7	0.080	0.084
Prospective	7	0.100	0.161
Prospective	19	0.080	0.086
Prospective	19	0.100	0.169
Prospective	53	0.080	0.127
Prospective	53	0.100	0.256
Prospective	100	0.080	0.122

Protocol	Test groups	Assumed gain	Power
Prospective	100	0.100	0.248
Prospective	400	0.080	0.136
Prospective	400	0.100	0.272

Table S13: Repeated source learning curves. Every non-full size uses twenty independently sampled subsets of the opened-source training pool; full-pool repeats vary learner seed but have one unique subset. Evaluation and calibration groups are fixed and independent of fitting. Standard deviations are across fits, not evidence from additional rivers.

Score	Protocol	Train groups	Subsets	Fits	Retention percent	SD pp
Generic + OOF	Retro	10	20	20	74.631	3.195
Direct H30	Retro	10	20	20	77.514	2.675
Generic H30	Retro	10	20	20	72.796	3.731
Window	Retro	10	20	20	77.495	2.388
Generic + OOF	Retro	25	20	20	77.619	1.557
Direct H30	Retro	25	20	20	78.981	1.881
Generic H30	Retro	25	20	20	76.417	1.675
Window	Retro	25	20	20	78.463	1.571
Generic + OOF	Retro	50	20	20	77.406	1.915
Direct H30	Retro	50	20	20	79.162	2.187
Generic H30	Retro	50	20	20	77.276	1.659
Window	Retro	50	20	20	79.334	1.981
Generic + OOF	Retro	75	20	20	77.860	0.750
Direct H30	Retro	75	20	20	79.541	0.840
Generic H30	Retro	75	20	20	79.009	1.195
Window	Retro	75	20	20	79.712	0.708
Generic + OOF	Retro	86	1	3	78.152	0.458
Direct H30	Retro	86	1	3	79.996	0.124
Generic H30	Retro	86	1	3	79.906	0.541

Score	Protocol	Train groups	Subsets	Fits	Retention percent	SD pp
Window	Retro	86	1	3	79.792	0.459
Generic + OOF	Prospective	10	20	20	72.627	2.837
Direct H30	Prospective	10	20	20	70.985	3.407
Generic H30	Prospective	10	20	20	69.027	5.188
Window	Prospective	10	20	20	71.321	3.455
Generic + OOF	Prospective	25	20	20	74.009	3.545
Direct H30	Prospective	25	20	20	74.054	1.954
Generic H30	Prospective	25	20	20	70.622	4.008
Window	Prospective	25	20	20	73.718	1.520
Generic + OOF	Prospective	50	20	20	74.667	1.786
Direct H30	Prospective	50	20	20	74.086	1.881
Generic H30	Prospective	50	20	20	71.285	2.883
Window	Prospective	50	20	20	73.835	2.349
Generic + OOF	Prospective	62	1	3	75.845	2.792
Direct H30	Prospective	62	1	3	74.882	2.476
Generic H30	Prospective	62	1	3	73.074	4.222
Window	Prospective	62	1	3	73.349	0.483

Table S14: Independent new-station transfer, with Swiss documented means primary and French project aggregates separate. F0: source model and source threshold unchanged; F1: same model, new local calibration; F2: new development fitting and local calibration; F0-P: target provider excluded; F0-US: USGS-only intercontinental source. A local reconstruction fit to permitted station history is allowed in every regime. The joint certificate delivers its own selected point; the other feature comparisons deliver the fixed information-priority point. None of these point risks is a per-network guarantee.

Score	Time	Provider	Mode	Keep percent	Fail percent	Groups
Generic H30	Retro	foen	F0	51.121	5.848	7
Generic H30	Retro	foen	F1	45.740	3.268	7
Generic H30	Retro	foen	F2	43.946	3.061	7
Complete diag.	Retro	foen	F0	53.214	5.899	7
Complete diag.	Retro	foen	F1	45.291	2.640	7
Complete diag.	Retro	foen	F2	41.106	1.818	7
Direct H30	Retro	foen	F0	56.353	9.284	7
Direct H30	Retro	foen	F1	46.487	2.894	7
Direct H30	Retro	foen	F2	45.291	3.960	7
Window	Retro	foen	F0	56.054	8.800	7
Window	Retro	foen	F1	47.534	4.088	7
Window	Retro	foen	F2	44.395	3.030	7
Joint certificate	Retro	foen	F0	52.915	10.169	7
Joint certificate	Retro	foen	F1	47.085	6.984	7
Joint certificate	Retro	foen	F2	25.262	1.183	7
Generic H30	Retro	foen	F0-P	54.709	8.470	7
Direct H30	Retro	foen	F0-P	60.389	11.386	7
Window	Retro	foen	F0-P	59.791	11.250	7
Joint certificate	Retro	foen	F0-P	49.327	7.879	7
Complete diag.	Retro	foen	F0-P	55.306	6.757	7
Generic H30	Retro	foen	F0-US	52.167	6.877	7

Score	Time	Provider	Mode	Keep percent	Fail percent	Groups
Direct H30	Retro	foen	F0-US	60.239	10.918	7
Window	Retro	foen	F0-US	56.951	9.711	7
Joint certificate	Retro	foen	F0-US	53.961	9.141	7
Complete diag.	Retro	foen	F0-US	55.605	6.989	7
Generic H30	Retro	hubeau	F0	48.857	1.462	12
Generic H30	Retro	hubeau	F1	59.143	3.623	12
Generic H30	Retro	hubeau	F2	55.000	3.377	12
Complete diag.	Retro	hubeau	F0	60.000	2.381	12
Complete diag.	Retro	hubeau	F1	52.714	1.355	12
Complete diag.	Retro	hubeau	F2	52.000	1.923	12
Direct H30	Retro	hubeau	F0	53.857	2.653	12
Direct H30	Retro	hubeau	F1	30.714	0.000	12
Direct H30	Retro	hubeau	F2	53.286	1.877	12
Window	Retro	hubeau	F0	69.143	6.198	12
Window	Retro	hubeau	F1	51.714	2.210	12
Window	Retro	hubeau	F2	46.286	2.160	12
Joint certificate	Retro	hubeau	F0	51.143	3.911	12
Joint certificate	Retro	hubeau	F1	18.429	0.000	12
Generic H30	Retro	hubeau	F0-P	47.571	1.502	12
Direct H30	Retro	hubeau	F0-P	54.286	2.895	12
Window	Retro	hubeau	F0-P	58.857	2.913	12
Joint certificate	Retro	hubeau	F0-P	55.000	5.195	12
Complete diag.	Retro	hubeau	F0-P	61.714	2.083	12
Generic H30	Retro	hubeau	F0-US	53.000	3.235	12
Direct H30	Retro	hubeau	F0-US	60.429	4.255	12
Window	Retro	hubeau	F0-US	61.429	4.651	12
Joint certificate	Retro	hubeau	F0-US	56.571	6.566	12
Complete diag.	Retro	hubeau	F0-US	63.571	2.921	12
Joint certificate	Retro	hubeau	F2	0.000	–	12
Generic H30	Prospective	foen	F0	43.353	6.667	7
Generic H30	Prospective	foen	F1	30.636	0.000	7

Score	Time	Provider	Mode	Keep percent	Fail percent	Groups
Generic H30	Prospective	foen	F2	30.058	1.923	7
Complete diag.	Prospective	foen	F0	46.243	6.250	7
Complete diag.	Prospective	foen	F1	32.948	0.000	7
Complete diag.	Prospective	foen	F2	32.948	0.000	7
Direct H30	Prospective	foen	F0	54.913	14.737	7
Direct H30	Prospective	foen	F1	34.104	0.000	7
Direct H30	Prospective	foen	F2	15.029	0.000	7
Window	Prospective	foen	F0	47.399	9.756	7
Window	Prospective	foen	F1	36.994	1.562	7
Window	Prospective	foen	F2	13.873	0.000	7
Joint certificate	Prospective	foen	F0	53.179	9.783	7
Joint certificate	Prospective	foen	F1	28.902	2.000	7
Joint certificate	Prospective	foen	F2	4.046	0.000	7
Generic H30	Prospective	foen	F0-P	42.197	5.479	7
Direct H30	Prospective	foen	F0-P	38.728	2.985	7
Window	Prospective	foen	F0-P	43.931	7.895	7
Joint certificate	Prospective	foen	F0-P	38.150	1.515	7
Complete diag.	Prospective	foen	F0-P	49.133	8.235	7
Generic H30	Prospective	foen	F0-US	43.353	6.667	7
Direct H30	Prospective	foen	F0-US	41.618	6.944	7
Window	Prospective	foen	F0-US	41.618	6.944	7
Joint certificate	Prospective	foen	F0-US	31.214	5.556	7
Complete diag.	Prospective	foen	F0-US	36.994	3.125	7
Generic H30	Prospective	hubeau	F0	35.294	0.000	6
Generic H30	Prospective	hubeau	F1	42.353	2.778	6
Generic H30	Prospective	hubeau	F2	36.471	0.000	6
Complete diag.	Prospective	hubeau	F0	40.000	5.882	6
Complete diag.	Prospective	hubeau	F1	40.000	5.882	6
Complete diag.	Prospective	hubeau	F2	42.353	0.000	6
Direct H30	Prospective	hubeau	F0	44.706	10.526	6
Direct H30	Prospective	hubeau	F1	37.647	6.250	6

Score	Time	Provider	Mode	Keep percent	Fail percent	Groups
Direct H30	Prospective	hubeau	F2	44.706	5.263	6
Window	Prospective	hubeau	F0	49.412	7.143	6
Window	Prospective	hubeau	F1	42.353	5.556	6
Window	Prospective	hubeau	F2	38.824	0.000	6
Joint certificate	Prospective	hubeau	F0	36.471	16.129	6
Joint certificate	Prospective	hubeau	F1	0.000	–	6
Generic H30	Prospective	hubeau	F0-P	45.882	7.692	6
Direct H30	Prospective	hubeau	F0-P	34.118	0.000	6
Window	Prospective	hubeau	F0-P	42.353	5.556	6
Joint certificate	Prospective	hubeau	F0-P	40.000	5.882	6
Complete diag.	Prospective	hubeau	F0-P	49.412	4.762	6
Generic H30	Prospective	hubeau	F0-US	43.529	5.405	6
Direct H30	Prospective	hubeau	F0-US	40.000	11.765	6
Window	Prospective	hubeau	F0-US	43.529	5.405	6
Joint certificate	Prospective	hubeau	F0-US	42.353	13.889	6
Complete diag.	Prospective	hubeau	F0-US	34.118	0.000	6
Joint certificate	Prospective	hubeau	F2	0.000	–	6

Table S15: Corrected hydrological strata for the uncertain-window score. Snow precipitation is a percent: majority snow means at least 50 percent, not 0.5. Glacier influence has priority. This descriptive unit correction changed no predictor, threshold, case, eligibility or frozen role. Single-component strata cannot certify their risk; raw script-360 hydrological labels are superseded by this table.

Hydrology	Time	Provider	Mode	Admit	Fail	Groups	Risk upper
glacier_influenced	Retro	foen	F0	68	16	1	–
other_mountain_catchment	Retro	foen	F0	159	9	3	0.086

Hydrology	Time	Provider	Mode	Admit	Fail	Groups	Risk upper
snow_dominated	Retro	foen	F0	148	8	3	0.100
glacier_influenced	Retro	foen	F1	60	9	1	–
other_mountain_catchment	Retro	foen	F1	137	3	3	0.050
snow_dominated	Retro	foen	F1	121	1	3	0.026
glacier_influenced	Retro	hubeau	F0	84	0	2	0.000
other_mountain_catchment	Retro	hubeau	F0	400	30	10	0.132
glacier_influenced	Retro	hubeau	F1	71	0	2	0.000
other_mountain_catchment	Retro	hubeau	F1	291	8	10	0.056
glacier_influenced	Prospective	foen	F0	19	1	1	–
other_mountain_catchment	Prospective	foen	F0	30	1	3	0.091
snow_dominated	Prospective	foen	F0	33	6	3	0.364
glacier_influenced	Prospective	foen	F1	18	0	1	–
other_mountain_catchment	Prospective	foen	F1	22	0	3	0.000
snow_dominated	Prospective	foen	F1	24	1	3	0.091
other_mountain_catchment	Prospective	hubeau	F0	42	3	6	0.167
other_mountain_catchment	Prospective	hubeau	F1	36	2	6	0.158

Table S16: Independent admission under warm anomalies, compound low-flow/heat overlap and the independently published event catalog. Flow missing on required gap-event days is explicitly unknown, not absence of compound stress. The project event threshold requires three days; published episodes require two. These are outcome strata, never deployment features.

Condition	Time	Provider	Mode	Admit	Fail	Groups	Risk upper
False	Retro	foen	F0	229	12	7	0.115
True	Retro	foen	F0	146	21	7	0.243
flow_incomplete_on_gap_event_days	Retro	foen	F0	13	3	7	0.400
gap_intersects_compound_event	Retro	foen	F0	58	17	7	0.489
no_gap_thermal_event_overlap	Retro	foen	F0	262	10	7	0.095
observed_no_compound_event_overlap	Retro	foen	F0	42	3	7	0.123
False	Retro	foen	F0	375	33	7	0.150
False	Retro	foen	F1	191	6	7	0.078
True	Retro	foen	F1	127	7	7	0.113
flow_incomplete_on_gap_event_days	Retro	foen	F1	11	1	7	0.231
gap_intersects_compound_event	Retro	foen	F1	42	7	7	0.381
no_gap_thermal_event_overlap	Retro	foen	F1	227	4	7	0.047
observed_no_compound_event_overlap	Retro	foen	F1	38	1	7	0.059
False	Retro	foen	F1	318	13	7	0.087
False	Retro	hubeau	F0	388	14	12	0.075
True	Retro	hubeau	F0	96	16	7	0.284

Condition	Time	Provider	Mode	Admit	Fail	Groups	Risk upper
flow_incomplete_on_gap_event_days	Retro	hubeau	F0	141	14	12	0.203
no_gap_thermal_event_overlap	Retro	hubeau	F0	343	16	12	0.091
False	Retro	hubeau	F0	484	30	12	0.115
False	Retro	hubeau	F1	286	3	12	0.031
True	Retro	hubeau	F1	76	5	7	0.123
flow_incomplete_on_gap_event_days	Retro	hubeau	F1	94	3	12	0.083
no_gap_thermal_event_overlap	Retro	hubeau	F1	268	5	12	0.049
False	Retro	hubeau	F1	362	8	12	0.049
False	Prospective	foen	F0	36	0	7	0.000
True	Prospective	foen	F0	46	8	7	0.304
flow_incomplete_on_gap_event_days	Prospective	foen	F0	10	2	7	0.333
gap_intersects_compound_event	Prospective	foen	F0	13	4	7	0.556
no_gap_thermal_event_overlap	Prospective	foen	F0	54	2	7	0.120
observed_no_compound_event_overlap	Prospective	foen	F0	5	0	6	0.000
False	Prospective	foen	F0	82	8	7	0.186
False	Prospective	foen	F1	32	0	7	0.000
True	Prospective	foen	F1	32	1	7	0.094
flow_incomplete_on_gap_event_days	Prospective	foen	F1	7	0	7	0.000
gap_intersects_compound_event	Prospective	foen	F1	7	1	7	0.273
no_gap_thermal_event_overlap	Prospective	foen	F1	46	0	7	0.000

Condition	Time	Provider	Mode	Admit	Fail	Groups	Risk upper
observed_no_compound_event_overlap	Prospective	foen	F1	4	0	6	0.000
False	Prospective	foen	F1	64	1	7	0.045
False	Prospective	hubeau	F0	18	1	4	0.176
True	Prospective	hubeau	F0	24	2	4	0.273
flow_incomplete_on_gap_event_days	Prospective	hubeau	F0	16	3	6	0.409
no_gap_thermal_event_overlap	Prospective	hubeau	F0	26	0	6	0.000
False	Prospective	hubeau	F0	42	3	6	0.167
False	Prospective	hubeau	F1	14	0	4	0.000
True	Prospective	hubeau	F1	22	2	4	0.300
flow_incomplete_on_gap_event_days	Prospective	hubeau	F1	12	2	6	0.364
no_gap_thermal_event_overlap	Prospective	hubeau	F1	24	0	6	0.000
False	Prospective	hubeau	F1	36	2	6	0.158

Table S17: Every independent new-station component for the uncertain-window policy. Counts expose concentrated failures that pooled risk can hide. Repeated masks and years within one river are not independent Bernoulli trials: a zero numerator is not a safety certificate and no case-binomial confidence interval is claimed.

Station	Time	Mode	Pool	Admit	Fail	Risk
CH_2347	Retro	F0	94	58	5	0.086
CH_2609	Retro	F0	94	45	2	0.044
CH_2179	Retro	F0	97	56	2	0.036
CH_2612	Retro	F0	97	47	3	0.064
CH_2604	Retro	F0	94	50	5	0.100

Station	Time	Mode	Pool	Admit	Fail	Risk
CH_2608	Retro	F0	97	51	0	0.000
CH_2327	Retro	F0	96	68	16	0.235
CH_2347	Retro	F1	94	49	1	0.020
CH_2609	Retro	F1	94	40	2	0.050
CH_2179	Retro	F1	97	48	0	0.000
CH_2612	Retro	F1	97	40	0	0.000
CH_2604	Retro	F1	94	39	1	0.026
CH_2608	Retro	F1	97	42	0	0.000
CH_2327	Retro	F1	96	60	9	0.150
MC_5099170	Retro	F0	67	52	1	0.019
PY_5180550	Retro	F0	58	46	4	0.087
MC_5096050	Retro	F0	59	46	0	0.000
PY_5180295	Retro	F0	88	60	4	0.067
AL_6078200	Retro	F0	55	32	6	0.188
PY_5218700	Retro	F0	24	24	0	0.000
AL_6152400	Retro	F0	57	29	9	0.310
PY_5200050	Retro	F0	44	36	0	0.000
PY_6175400	Retro	F0	51	29	5	0.172
PY_5172350	Retro	F0	71	49	1	0.020
AL_6143650	Retro	F0	69	60	0	0.000
PY_5110000	Retro	F0	57	21	0	0.000
MC_5099170	Retro	F1	67	42	0	0.000
PY_5180550	Retro	F1	58	37	1	0.027
MC_5096050	Retro	F1	59	37	0	0.000
PY_5180295	Retro	F1	88	26	0	0.000
AL_6078200	Retro	F1	55	25	2	0.080
PY_5218700	Retro	F1	24	20	0	0.000
AL_6152400	Retro	F1	57	19	2	0.105
PY_5200050	Retro	F1	44	24	0	0.000
PY_6175400	Retro	F1	51	23	3	0.130
PY_5172350	Retro	F1	71	39	0	0.000
AL_6143650	Retro	F1	69	51	0	0.000

Station	Time	Mode	Pool	Admit	Fail	Risk
PY_5110000	Retro	F1	57	19	0	0.000
CH_2347	Prospective	F0	26	11	0	0.000
CH_2609	Prospective	F0	24	8	0	0.000
CH_2179	Prospective	F0	25	11	1	0.091
CH_2612	Prospective	F0	25	11	4	0.364
CH_2604	Prospective	F0	24	13	2	0.154
CH_2608	Prospective	F0	25	9	0	0.000
CH_2327	Prospective	F0	24	19	1	0.053
CH_2347	Prospective	F1	26	8	0	0.000
CH_2609	Prospective	F1	24	6	0	0.000
CH_2179	Prospective	F1	25	8	0	0.000
CH_2612	Prospective	F1	25	6	0	0.000
CH_2604	Prospective	F1	24	11	1	0.091
CH_2608	Prospective	F1	25	7	0	0.000
CH_2327	Prospective	F1	24	18	0	0.000
MC_5099170	Prospective	F0	11	7	0	0.000
PY_5180550	Prospective	F0	6	5	0	0.000
PY_5180295	Prospective	F0	12	4	1	0.250
AL_6078200	Prospective	F0	26	13	0	0.000
AL_6152400	Prospective	F0	24	10	2	0.200
PY_5172350	Prospective	F0	6	3	0	0.000
MC_5099170	Prospective	F1	11	5	0	0.000
PY_5180550	Prospective	F1	6	5	0	0.000
PY_5180295	Prospective	F1	12	3	0	0.000
AL_6078200	Prospective	F1	26	12	0	0.000
AL_6152400	Prospective	F1	24	9	2	0.222
PY_5172350	Prospective	F1	6	2	0	0.000

Table S18: Constructed population stress test, not observed river failure rates. Groups are independent with perfectly dependent failures within groups, true failure probability 0.06 and uninformative scores. Empty admissions are not zero risk. The rigorous bounded-group test addresses a group-expectation ratio under its assumptions, not every catchment or arbitrary domain shift.

scenario	n repetitions	empirical violation probability	rigorous violation probability	empirical admission probability	rigorous admission probability
forty_perfectly_correlated	1000	0.085	0.000	0.085	0.000
hundred_perfectly_correlated	1000	0.047	0.002	0.047	0.002
rare_bad_group	1000	0.000	0.000	0.450	0.000
twelve_perfectly_correlated	1000	0.454	0.000	0.454	0.000

Table S19: Opened-source component-held-out event/cofailure assay. Exactly the admission-trained heads and thresholds are applied without adaptation. Values average the three dependent split sensitivities. Before/during/after masks are relative to the same longest training-threshold episode. The new European component bundles are single-station and cannot by themselves identify donor-cofailure sensitivity; this nondegenerate multi-station test supplies that diagnostic, not independent confirmation.

Gap phase	Inputs	Removed share	Retention	Selected risk	Candidates
after	Retro	0.000	0.858	0.025	7938
before	Retro	0.000	0.864	0.025	6932
during	Retro	0.000	0.766	0.062	7017
after	Retro	0.500	0.837	0.024	7938
before	Retro	0.500	0.839	0.027	6932

Gap phase	Inputs	Removed share	Retention	Selected risk	Candidates
during	Retro	0.500	0.739	0.072	7017
after	Retro	1.000	0.764	0.038	7938
before	Retro	1.000	0.780	0.034	6932
during	Retro	1.000	0.663	0.112	7017
after	Past-local	0.000	0.846	0.019	7323
before	Past-local	0.000	0.858	0.013	6141
during	Past-local	0.000	0.763	0.051	6482
after	Past-local	0.500	0.824	0.018	7323
before	Past-local	0.500	0.827	0.017	6141
during	Past-local	0.500	0.724	0.055	6482
after	Past-local	1.000	0.752	0.025	7323
before	Past-local	1.000	0.774	0.023	6141
during	Past-local	1.000	0.654	0.092	6482

Table S20: No-screen composition rule on independent natural-calendar replay. Only French records contribute qualifying calendars here; there is no independent Swiss documented-mean calendar estimate. Bin boundaries were inherited from opened analyses; cell risks were fitted on opened-source empirical geometries and fixed before new-test replay. Advice based on a point forecast is not certification; source-only prediction intervals omit unmodelled domain shift and one target component cannot establish safety.

Component	Time	Masks	Station-years	Predicted risk	Observed risk	No-screen advice
External 1	Retro	30	6	0.286	0.133	False
External 2	Retro	18	9	0.079	0.000	False
External 3	Retro	24	8	0.415	0.333	False
External 4	Retro	8	4	0.276	0.000	False
External 5	Retro	20	5	0.415	0.700	False
External 6	Retro	8	4	0.415	0.125	False

Component	Time	Masks	Station-years	Predicted risk	Observed risk	No-screen advice
External 7	Retro	16	8	0.218	0.500	False
External 8	Retro	22	11	0.015	0.000	True
External 9	Retro	12	6	0.415	0.000	False
External 10	Retro	10	5	0.215	0.100	False
External 1	Prospective	3	1	0.257	0.000	False
External 2	Prospective	2	1	0.149	0.000	False
External 3	Prospective	3	1	0.475	0.667	False
External 5	Prospective	8	2	0.475	0.500	False
External 8	Prospective	2	1	0.040	0.000	True

Table S21: Reconstruction references on identical replicate-zero crossed cases. Availability is explicit, and the fixed policy MAE is recomputed on each method's identical supported subset. Inputs differ, so differences are not attributed solely to architecture. CSDI is an old-source-trained, fixed eight-draw retrospective univariate reference, not a claim to optimized state of the art. Air2stream parameters and discharge normalization exclude the target year and are strictly prior prospectively; every gap starts from an observed left anchor. No unsupported prediction is counted as a success.

Reconstructor	Time	Information	Available	H30 MAE C	Matched policy C	Failure share
CSDI: median series	Retro	Target only	1701	0.453	0.311	0.304
CSDI: median H30	Retro	Target only	1701	0.434	0.311	0.290
Daily climate	Retro	Target only	1701	0.401	0.311	0.297
Smooth climate	Retro	Target only	1701	0.399	0.311	0.305

Reconstructor	Time	Information	Available	H30 MAE C	Matched policy C	Failure share
Interpolation	Retro	Target only	1701	0.428	0.311	0.273
Neighbour ridge	Retro	Target + neighbours	0	–	–	–
Air ridge	Retro	Target + air	1544	0.280	0.280	0.215
air2stream3	Retro	Target + air	1544	0.254	0.280	0.192
air2stream8	Retro	Target + air + flow	821	0.203	0.342	0.157
air2stream3	Prospective	Target + air	185	0.277	0.332	0.211
air2stream8	Prospective	Target + air + flow	135	0.173	0.368	0.104

Table S22: Raw 90 percent conditional-trajectory band diagnostics. These are uncalibrated scenario bands for the fixed priority point, not valid confidence intervals or selected-risk certificates. Missing bands remain missing. The conditional residual bootstrap does not capture all bias or transfer uncertainty.

Assay	Time	Provider	Candidates	Bands	Coverage	Width C
admission	Retro	foen	669	669	0.716	0.798
admission	Retro	hubeau	700	700	0.813	0.546
crossed	Retro	foen	6718	6718	0.730	0.770
crossed	Retro	hubeau	5631	5631	0.831	0.535
events	Retro	foen	582	582	0.753	0.374
events	Retro	hubeau	492	492	0.904	0.326
real_calendars	Retro	hubeau	168	168	0.804	0.738
admission	Prospective	foen	173	173	0.607	0.943

Assay	Time	Provider	Candidates	Bands	Coverage	Width C
admission	Prospective	hubeau	85	85	0.682	0.780
crossed	Prospective	foen	1790	1790	0.684	0.895
crossed	Prospective	hubeau	655	655	0.794	0.710
events	Prospective	foen	182	182	0.780	0.349
events	Prospective	hubeau	67	67	0.806	0.385
real_calendars	Prospective	hubeau	18	18	0.667	1.001

Table S23: Full original-candidate locally anomalous-hot decisions, fixed source uncertain-window policy. False-hot cost is one; costs were declared before new test. Negative differences favor screening. Always-unknown cost equals the declared abstention cost, with no magnitude estimate or finite selected risk. This stronger deterministic comparator was added in the post-test scientific audit without changing any decision. Always-hot/cool costs and the original descriptive component-bootstrap intervals remain in the source CSV; no simultaneous guarantee or measured agency preference is claimed.

Time	Provider	Miss cost	Unknown cost	Screen loss	vs all recon.	vs all unknown
Retro	foen	1.000	0.000	0.019	-0.072	0.019
Retro	foen	1.000	0.010	0.024	-0.067	0.014
Retro	foen	1.000	0.025	0.030	-0.061	0.005
Retro	foen	1.000	0.050	0.041	-0.050	-0.009
Retro	foen	1.000	0.100	0.063	-0.028	-0.037
Retro	foen	1.000	0.250	0.129	0.038	-0.121
Retro	foen	5.000	0.000	0.097	-0.359	0.097
Retro	foen	5.000	0.010	0.102	-0.354	0.092
Retro	foen	5.000	0.025	0.108	-0.348	0.083
Retro	foen	5.000	0.050	0.119	-0.337	0.069
Retro	foen	5.000	0.100	0.141	-0.315	0.041
Retro	foen	5.000	0.250	0.207	-0.249	-0.043

Time	Provider	Miss cost	Unknown cost	Screen loss	vs all recon.	vs all unknown
Retro	foen	10.000	0.000	0.194	-0.717	0.194
Retro	foen	10.000	0.010	0.199	-0.713	0.189
Retro	foen	10.000	0.025	0.205	-0.707	0.180
Retro	foen	10.000	0.050	0.216	-0.696	0.166
Retro	foen	10.000	0.100	0.238	-0.674	0.138
Retro	foen	10.000	0.250	0.304	-0.608	0.054
Retro	hubeau	1.000	0.000	0.060	-0.079	0.060
Retro	hubeau	1.000	0.010	0.063	-0.075	0.053
Retro	hubeau	1.000	0.025	0.068	-0.071	0.043
Retro	hubeau	1.000	0.050	0.075	-0.063	0.025
Retro	hubeau	1.000	0.100	0.091	-0.048	-0.009
Retro	hubeau	1.000	0.250	0.137	-0.001	-0.113
Retro	hubeau	5.000	0.000	0.203	-0.210	0.203
Retro	hubeau	5.000	0.010	0.206	-0.207	0.196
Retro	hubeau	5.000	0.025	0.211	-0.202	0.186
Retro	hubeau	5.000	0.050	0.218	-0.195	0.168
Retro	hubeau	5.000	0.100	0.234	-0.179	0.134
Retro	hubeau	5.000	0.250	0.280	-0.133	0.030
Retro	hubeau	10.000	0.000	0.381	-0.374	0.381
Retro	hubeau	10.000	0.010	0.385	-0.371	0.375
Retro	hubeau	10.000	0.025	0.389	-0.367	0.364
Retro	hubeau	10.000	0.050	0.397	-0.359	0.347
Retro	hubeau	10.000	0.100	0.412	-0.343	0.312
Retro	hubeau	10.000	0.250	0.459	-0.297	0.209
Prospective	foen	1.000	0.000	0.012	-0.098	0.012
Prospective	foen	1.000	0.010	0.017	-0.093	0.007
Prospective	foen	1.000	0.025	0.025	-0.085	-0.000
Prospective	foen	1.000	0.050	0.038	-0.072	-0.012
Prospective	foen	1.000	0.100	0.064	-0.046	-0.036
Prospective	foen	1.000	0.250	0.143	0.033	-0.107
Prospective	foen	5.000	0.000	0.058	-0.491	0.058

Time	Provider	Miss cost	Unknown cost	Screen loss	vs all recon.	vs all unknown
Prospective	foen	5.000	0.010	0.063	-0.486	0.053
Prospective	foen	5.000	0.025	0.071	-0.478	0.046
Prospective	foen	5.000	0.050	0.084	-0.465	0.034
Prospective	foen	5.000	0.100	0.110	-0.439	0.010
Prospective	foen	5.000	0.250	0.189	-0.360	-0.061
Prospective	foen	10.000	0.000	0.116	-0.983	0.116
Prospective	foen	10.000	0.010	0.121	-0.977	0.111
Prospective	foen	10.000	0.025	0.129	-0.970	0.104
Prospective	foen	10.000	0.050	0.142	-0.956	0.092
Prospective	foen	10.000	0.100	0.168	-0.930	0.068
Prospective	foen	10.000	0.250	0.247	-0.851	-0.003
Prospective	hubeau	1.000	0.000	0.024	-0.188	0.024
Prospective	hubeau	1.000	0.010	0.029	-0.183	0.019
Prospective	hubeau	1.000	0.025	0.036	-0.176	0.011
Prospective	hubeau	1.000	0.050	0.049	-0.163	-0.001
Prospective	hubeau	1.000	0.100	0.074	-0.138	-0.026
Prospective	hubeau	1.000	0.250	0.150	-0.062	-0.100
Prospective	hubeau	5.000	0.000	0.024	-0.565	0.024
Prospective	hubeau	5.000	0.010	0.029	-0.560	0.019
Prospective	hubeau	5.000	0.025	0.036	-0.552	0.011
Prospective	hubeau	5.000	0.050	0.049	-0.539	-0.001
Prospective	hubeau	5.000	0.100	0.074	-0.514	-0.026
Prospective	hubeau	5.000	0.250	0.150	-0.438	-0.100
Prospective	hubeau	10.000	0.000	0.024	-1.035	0.024
Prospective	hubeau	10.000	0.010	0.029	-1.030	0.019
Prospective	hubeau	10.000	0.025	0.036	-1.023	0.011
Prospective	hubeau	10.000	0.050	0.049	-1.010	-0.001
Prospective	hubeau	10.000	0.100	0.074	-0.985	-0.026
Prospective	hubeau	10.000	0.250	0.150	-0.909	-0.100

Table S24: Locally hot and cool refusals, selected-sample frequency error and original-pool unknown-label bounds. Bounds allow every unknown label to be cool or hot and are not statistical confidence intervals; classification mistakes among judged cases can still affect them. Magnitude and hot-label admission are distinct outputs. Declaring an observed hot lower bound never certifies a cool record.

Time	Provider	Decision	Hot refusal	Cool refusal	Freq. error	Pool lower	Pool upper
Retro	foen	Magnitude	0.464	0.434	-0.042	0.070	0.510
Retro	foen	Hot-label rescue	0.446	0.434	-0.037	0.073	0.510
Retro	hubeau	Magnitude	0.355	0.296	-0.031	0.129	0.437
Retro	hubeau	Hot-label rescue	0.349	0.296	-0.030	0.130	0.437
Prospective	foen	Magnitude	0.488	0.538	-0.005	0.116	0.642
Prospective	foen	Hot-label rescue	0.465	0.538	0.004	0.121	0.642
Prospective	hubeau	Magnitude	0.400	0.580	0.136	0.271	0.776
Prospective	hubeau	Hot-label rescue	0.371	0.580	0.146	0.282	0.776

S10 Supplementary objective and use comparisons

The four-cell supplement holds the H30 point and generic-plus-direct-uncertainty information fixed while crossing quantile/error-event learning with absence/presence of seven window summaries. Quantile regression and binary log-loss boosting share 350 iterations, depth four, learning rate 0.06 and leaf size forty. Identical group assignments, learner seeds, candidate construction and calibration resamples reduce extraneous differences. Original confirmed hypotheses and source/target locks are preserved. Five outer folds for three seeds give one held-out prediction per candidate per seed. Fixed-retention comparisons rank within each held-out fold with deterministic identity-based tie breaking; evaluation labels never select the rank.

Direct hot-probability models add only the available point and threshold margin. Platt fitting uses calibration groups separate from learner fitting and testing. A one-class calibration set uses a smoothed constant rather than certainty. Source and local calibration are separate policies; calibration quality is measured and no guarantee of conditional probability is presumed. Every decision compares cool, hot and unknown conditional losses. Absolute loss, missed-hot, false-hot and unknown counts use the original candidate denominator. Uniform random rejection is an analytic expectation. Neither a favorable cost cell nor source cross-fitting is independent confirmation after review.

The original Swiss F0 screen and the supplementary direct-probability baseline share the 669 retrospective or 173 forward candidates. Their probability source heads use the original independent 80/20 source group roles; forward fitting/calibration targets stop at 2018. Probability fitting has no access to target test labels. Current target outcomes were already opened before this supplementary design, so the resulting comparison remains development evidence even with group-disjoint fitting.

Whole-procedure resampling preserves each station's local reconstruction, since resampling a complete component changes no local inputs, but refits all four global heads and thresholds. Evaluation components are resampled too. Source-like resampling does not manufacture a new failure type or validate exchangeability under regional climate dependence. The regional forward-event assay adds a stronger holdout on the score: target-provider components are excluded, global target years stop at 2014, and event evaluations use 2015, 2018 and 2022. Provider proxies do not equal measured process classes.

S11 Mechanism, trajectories and the next independent study

The exact-hidden 30/90-day length subset has 396 distinct station-years represented by 400 network-target-year rows. The original documented-mean mechanism pool has 4,720 distinct station-years represented by 4,884 such rows. Shared rows remain clustered in their complete record components. Common-subset contrasts accompany the original population. In the added nested assay, the true maximizing thirty-day interval defines a hidden seed, extended left, right or symmetrically to

sixty and ninety days. A thirty-day nonoverlapping control is displaced ninety days in the first feasible predetermined direction. All method contrasts use the same feasible years within direction, retaining boundary distances. These true-window designs identify sensitivity in an artificial mechanism assay and cannot be used as deployment rules.

The experimental state trajectory model separates a sixty-day smoothed historical residual path from shorter residuals. Historical-year analogs are weighted by visible water residuals and available air/flow anomalies relative to permitted monthly training history; a ten-percent uniform mixture preserves some analog diversity. Fourteen-day short blocks are matched within thirty seasonal days and never cross years. Historical residuals exclude the target year and all future years prospectively. Observations remain fixed. Two usable residual years are required, with unsupported cases explicit. Original raw blocks, visible correction and this candidate are compared on their common support, including CRPS, band width/coverage, hot-probability Brier score and warm/low-flow conditions. Nested samples of 32, 128 and 512 quantify Monte Carlo sensitivity. More draws cannot correct an incorrect residual distribution.

The reconstruction factorial refits each method's own numerical-error event on development groups and calibrates on the separate calibration role before evaluating the original test role. Fitting/calibration retain exact shared cases within information tiers. Both paired supported-subset results and full original-pool decisions are reported; unavailable reconstructions remain unknown in the latter, with their cost charged. Shared generic/window features are supplied to every method, without passing another model's direct uncertainty off as its own. The Air2stream-8 tier adds flow; fixed univariate eight-draw CSDI is a limited reference, not the best attainable probabilistic model. A future probability-model selection budget uses independent development groups and H30 CRPS, rather than selecting configurations on this opened test.

History sensitivity retains only target years feasible for all three budgets, then uses three, five or ten permitted complete years. Retrospective histories are closest in calendar distance with fixed tie breaking; forward histories are latest strictly earlier years. Start/end-of-year thirty- and ninety-day gaps, a fixed interior gap and

Table S25. Four-cell objective/window analysis on the identical documented-mean admission pool. Ranges span three split seeds, not independent replications. Q is the 0.9 absolute-error quantile; P is direct failure probability. Retention and selected risk are percentages. Formal admission is zero throughout.

protocol	arm	N	G	ret_min	ret_max	risk_min	risk_max
Retro	P direct	54837	144	77.491	78.680	2.530	2.935
Retro	P window	54837	144	77.546	78.781	2.469	2.813
Retro	Q direct	54837	144	77.676	79.933	2.430	3.082
Retro	Q window	54837	144	77.745	79.249	2.418	2.837
Past local	P direct	36471	144	74.160	75.249	2.199	2.419
Past local	P window	36471	144	75.219	76.485	2.369	2.864
Past local	Q direct	36471	144	75.679	76.518	2.667	2.839
Past local	Q window	36471	144	76.302	76.845	2.702	2.797

two fragments expose cases omitted by the original Swiss admission generator. The long-history subset does not replace short-history primary records or identify natural outage frequencies.

The next-study specification fixes archival H30 magnitude as the primary use. Joint success requires risk evidence at or below the declared cap and lower paired retention-gain bounds above five points against both same-information direct references: native conditional-CDF failure probability and matched failure-probability boosting. It separates artificial mechanism masks, actual-calendar replay and natural gaps recoverable from independent backups. New stations, process labels, event relations and local calibration budgets must be established before outcome opening. Real backup truth and a feasible costly data action are not currently supplied by synthetic masking. The acquisition branch remains inactive until those actions and costs exist; free available inputs are incorporated directly. A failure of the joint endpoint stops the core superiority claim without redefining the population or cost grid. The prospective specification is in `configs/wrr_next_confirmation_20261021.yaml`; it is not an executed new confirmation.

S11.1 Expanded probability development and natural-record readiness

The developed neural reference conditions on water and available air, unlike the original fixed univariate reference. Both new widths share data and development limits. It is therefore an information and budget change relative to the original refer-

Table S26. Complete-procedure refits of all four cells. Learning, calibration groups and evaluation groups are resampled. Retention differences are percentage points. The last column requires both a five-point sample gain and the specified test-group bounded-risk evidence; it is a descriptive bootstrap frequency, not attained power in a new population.

Protocol	Comparison	Refits	Difference pp	Lower	Upper	Joint fraction
Retro	Q window minus Q direct	1000	0.354	-4.795	5.433	0.000
Retro	P window minus P direct	1000	0.697	-4.703	5.820	0.000
Retro	P direct minus Q direct	1000	-1.134	-6.693	5.055	0.000
Retro	P window minus Q window	1000	-0.791	-6.497	5.148	0.000
Forward	Q window minus Q direct	1000	0.365	-5.089	5.759	0.000
Forward	P window minus P direct	1000	0.075	-5.607	5.423	0.000
Forward	P direct minus Q direct	1000	-2.016	-13.141	5.672	0.000
Forward	P window minus Q window	1000	-2.306	-12.851	5.012	0.000

Table S27. Use-matched Swiss decisions at missed-hot cost five and unknown cost 0.05. The original screen is unchanged; added probability policies are post-test development. Random-matched counts are exact expectations. All other counts are realized outcomes in the same original pool.

Protocol	arm	N	loss	FN	FP	Unknown
Retro	All reconstructed	669	0.456	61.000	0.000	0
Retro	Frozen screen	669	0.119	13.000	0.000	294
Retro	All unknown	669	0.050	0.000	0.000	669
Retro	All hot	669	0.833	0.000	557.000	0
Retro	All cool	669	0.837	112.000	0.000	0
Retro	Source probability	669	0.075	8.000	0.000	206
Retro	Local probability	669	0.059	5.000	0.000	283
Retro	Source prob. + window	669	0.067	7.000	0.000	202
Retro	Local prob. + window	669	0.064	6.000	0.000	254
Retro	Matched random	669	0.278	34.193	0.000	294
Forward	All reconstructed	173	0.549	19.000	0.000	0
Forward	Frozen screen	173	0.084	2.000	0.000	91
Forward	All unknown	173	0.050	0.000	0.000	173
Forward	All hot	173	0.751	0.000	130.000	0
Forward	All cool	173	1.243	43.000	0.000	0
Forward	Source probability	173	0.029	0.000	0.000	100
Forward	Local probability	173	0.036	0.000	0.000	124
Forward	Source prob. + window	173	0.028	0.000	0.000	98
Forward	Local prob. + window	173	0.035	0.000	0.000	120
Forward	Matched random	173	0.287	9.006	0.000	91

Table S28. Every Swiss station in the original frozen-window policy. No high-risk station is removed from the primary result. Detailed accepted-failure years, gap positions, proxy residuals and flow coverage remain in the accompanying case table.

station_id	N	Years	admitted	Failed	Risk	Bias C	MAE C
CH_2179	97	15	56	2	0.036	-0.191	0.344
CH_2327	96	15	68	16	0.235	-0.445	0.446
CH_2347	94	15	58	5	0.086	-0.376	0.412
CH_2604	94	15	50	5	0.100	-0.379	0.477
CH_2608	97	15	51	0	0.000	-0.256	0.316
CH_2609	94	15	45	2	0.044	-0.296	0.595
CH_2612	97	15	47	3	0.064	-0.470	0.601

Table S29. Information intervention with identical masks and recomputed reconstruction/uncertainty. Entries average three source splits for the full quantile model. Fitting and calibration information vary independently. This separates one information channel from domain shift without identifying its full cause.

fitting_information	calibration_information	population	Retention	Risk	Protocol
No donors	No donors	Swiss test	0.523	0.056	Retro
No donors	No donors	Source no donor	0.699	0.030	Retro
No donors	No donors	Source original	0.726	0.019	Retro
No donors	Original	Swiss test	0.523	0.056	Retro
No donors	Original	Source no donor	0.699	0.030	Retro
No donors	Original	Source original	0.726	0.019	Retro
Original	No donors	Swiss test	0.500	0.046	Retro
Original	No donors	Source no donor	0.671	0.024	Retro
Original	No donors	Source original	0.731	0.013	Retro
Original	Original	Swiss test	0.570	0.091	Retro
Original	Original	Source no donor	0.720	0.040	Retro
Original	Original	Source original	0.779	0.024	Retro
No donors	No donors	Swiss test	0.462	0.079	Past local
No donors	No donors	Source no donor	0.664	0.021	Past local
No donors	No donors	Source original	0.677	0.012	Past local
No donors	Original	Swiss test	0.503	0.091	Past local
No donors	Original	Source no donor	0.692	0.030	Past local
No donors	Original	Source original	0.715	0.018	Past local
Original	No donors	Swiss test	0.422	0.048	Past local
Original	No donors	Source no donor	0.629	0.014	Past local
Original	No donors	Source original	0.693	0.007	Past local
Original	Original	Swiss test	0.476	0.097	Past local
Original	Original	Source no donor	0.677	0.029	Past local
Original	Original	Source original	0.743	0.015	Past local

Table S30. Truth-assisted nested-gap mechanism assay. Every direction compares length and position in its own common feasible station-years. True support stays fully hidden under extension; the thirty-day control is displaced ninety days. These are mechanism masks, not deployable peak-location rules.

Protocol	method	direction	contrast	mean	ci_low	ci_high	G	Years
Retro	Climatology	left	Length 90-30	0.103	0.087	0.117	144	4720
Retro	Climatology	left	Position full-none	0.818	0.742	0.887	144	4720
Retro	Climatology	balanced	Length 90-30	0.171	0.152	0.189	144	4714
Retro	Climatology	balanced	Position full-none	0.818	0.742	0.888	144	4714
Retro	Climatology	right	Length 90-30	0.067	0.057	0.077	144	4690
Retro	Climatology	right	Position full-none	0.820	0.743	0.889	144	4690
Retro	Interpolation	left	Length 90-30	0.465	0.442	0.491	144	4720
Retro	Interpolation	left	Position full-none	0.790	0.740	0.836	144	4720
Retro	Interpolation	balanced	Length 90-30	1.527	1.417	1.655	144	4713
Retro	Interpolation	balanced	Position full-none	0.790	0.740	0.837	144	4713
Retro	Interpolation	right	Length 90-30	0.582	0.546	0.615	144	4689
Retro	Interpolation	right	Position full-none	0.789	0.738	0.836	144	4689
Past local	Climatology	left	Length 90-30	0.119	0.092	0.145	144	3388
Past local	Climatology	left	Position full-none	0.893	0.779	0.990	144	3388
Past local	Climatology	balanced	Length 90-30	0.201	0.172	0.227	144	3384
Past local	Climatology	balanced	Position full-none	0.894	0.780	0.990	144	3384
Past local	Climatology	right	Length 90-30	0.073	0.062	0.086	144	3366
Past local	Climatology	right	Position full-none	0.896	0.781	0.992	144	3366
Past local	Interpolation	left	Length 90-30	0.463	0.437	0.491	144	3388
Past local	Interpolation	left	Position full-none	0.770	0.716	0.817	144	3388
Past local	Interpolation	balanced	Length 90-30	1.535	1.417	1.669	144	3384
Past local	Interpolation	balanced	Position full-none	0.771	0.717	0.818	144	3384
Past local	Interpolation	right	Length 90-30	0.577	0.533	0.617	144	3365
Past local	Interpolation	right	Position full-none	0.769	0.715	0.817	144	3365

Table S31. Trajectory quality at 128 draws on each role's intersection of supported cases. Unsupported state-model cases are reported separately and never treated as zero uncertainty. State conditioning uses visible water residuals and available air/flow states; common-subset performance is supplementary development, not new confirmation.

role	variant	draws	N	G	MAE_C	bias_C	coverage90	width_C	CRPS_C	Brier
development	Raw blocks	128	1426	11	0.277	-0.101	0.804	0.591	0.210	0.057
development	State residuals	128	1426	11	0.221	-0.074	0.899	0.763	0.157	0.041
development	Visible correction	128	1426	11	0.263	-0.107	0.811	0.599	0.198	0.056
calibration	Raw blocks	128	1115	9	0.266	-0.119	0.762	0.464	0.206	0.052
calibration	State residuals	128	1115	9	0.192	-0.057	0.913	0.698	0.137	0.039
calibration	Visible correction	128	1115	9	0.251	-0.130	0.763	0.464	0.192	0.046
test	Raw blocks	128	907	7	0.362	-0.202	0.734	0.731	0.272	0.063
test	State residuals	128	907	7	0.272	-0.084	0.896	0.976	0.189	0.048
test	Visible correction	128	907	7	0.342	-0.226	0.749	0.736	0.253	0.058

Table S32. Reconstruction–admission factorial on all original Swiss controlled candidates in seven components, with unavailable reconstructions retained as unknown. Fitting/calibration use the identical shared subset within each information tier; matched reconstruction errors and paired supported-subset results are supplied in the source CSVs. Each method learns its own error event. Window features come from the common reconstruction for every method. CSDI retains its fixed eight-draw scope, not its performance ceiling; all formal policies refuse everything.

tier	method	window	N	Avail.	Unavail.	Accepted	Failed	Retention	Risk
Air subset	Priority	False	945	945	0	241	2	0.255	0.008
Air subset	Priority	True	945	945	0	278	0	0.294	0.000
Air subset	Air2stream-3	False	945	828	117	159	0	0.168	0.000
Air subset	Air2stream-3	True	945	828	117	334	2	0.353	0.006
Air subset	CSDI (8)	False	945	945	0	359	21	0.380	0.058
Air subset	CSDI (8)	True	945	945	0	399	10	0.422	0.025
Flow subset	Priority	False	945	945	0	209	1	0.221	0.005
Flow subset	Priority	True	945	945	0	162	0	0.171	0.000
Flow subset	Air2stream-8	False	945	821	124	387	15	0.410	0.039
Flow subset	Air2stream-8	True	945	821	124	405	10	0.429	0.025
Flow subset	State residuals	False	945	907	38	193	10	0.204	0.052
Flow subset	State residuals	True	945	907	38	421	16	0.446	0.038
Flow subset	Raw blocks	False	945	945	0	263	12	0.278	0.046
Flow subset	Raw blocks	True	945	945	0	257	2	0.272	0.008
Flow subset	Visible correction	False	945	945	0	151	3	0.160	0.020
Flow subset	Visible correction	True	945	945	0	198	3	0.210	0.015

Table S33. History length and explicit year-boundary/fragmented gaps on identical target years feasible for all three history budgets. These long-history comparison subsets do not replace the original population. Prospective histories are strictly earlier; retrospective histories use the closest permitted years.

history_years	design	n	groups	available	mae_C	failure_fraction	Protocol
3	end30	270	18	270	0.000	0.000	Retro
3	end90	270	18	270	0.000	0.000	Retro
3	fragmented60	270	18	270	0.321	0.252	Retro
3	interior60	270	18	270	0.780	0.507	Retro
3	start30	270	18	270	0.000	0.000	Retro
3	start90	270	18	270	0.000	0.000	Retro
5	end30	270	18	270	0.000	0.000	Retro
5	end90	270	18	270	0.000	0.000	Retro
5	fragmented60	270	18	270	0.315	0.241	Retro
5	interior60	270	18	270	0.768	0.519	Retro
5	start30	270	18	270	0.000	0.000	Retro
5	start90	270	18	270	0.000	0.000	Retro
10	end30	270	18	270	0.000	0.000	Retro
10	end90	270	18	270	0.000	0.000	Retro
10	fragmented60	270	18	270	0.311	0.226	Retro
10	interior60	270	18	270	0.743	0.522	Retro
10	start30	270	18	270	0.000	0.000	Retro
10	start90	270	18	270	0.000	0.000	Retro
3	end30	72	18	72	0.000	0.000	Forward
3	end90	72	18	72	0.000	0.000	Forward
3	fragmented60	72	18	72	0.232	0.153	Forward
3	interior60	72	18	72	1.203	0.667	Forward
3	start30	72	18	72	0.000	0.000	Forward
3	start90	72	18	72	0.000	0.000	Forward
5	end30	72	18	72	0.000	0.000	Forward
5	end90	72	18	72	0.000	0.000	Forward
5	fragmented60	72	18	72	0.247	0.153	Forward
5	interior60	72	18	72	1.066	0.667	Forward
5	start30	72	18	72	0.000	0.000	Forward
5	start90	72	18	72	0.000	0.000	Forward
10	end30	72	18	72	0.000	0.000	Forward
10	end90	72	18	72	0.000	0.000	Forward
10	fragmented60	72	18	72	0.281	0.208	Forward
10	interior60	72	18	72	1.107	0.736	Forward
10	start30	72	18	72	0.000	0.000	Forward
10	start90	72	18	72	0.000	0.000	Forward

Table S34. Target calibration budget sensitivity for the original frozen score on the unchanged Swiss test pool. Budgets count independent calibration groups. Means and ranges span enumerated or fixed-seed sampled subsets, not new independent test repetitions. No subset certifies positive admission under the formal rule.

Cal. G	subsets	retention_mean	retention_min	retention_max	risk_mean	risk_min	risk_max	Protocol
1	9	0.654	0.420	1.000	0.144	0.014	0.342	Retro
3	30	0.463	0.420	0.561	0.035	0.014	0.088	Retro
5	30	0.448	0.420	0.475	0.028	0.014	0.041	Retro
9	1	0.475	0.475	0.475	0.041	0.041	0.041	Retro
1	9	0.553	0.324	1.000	0.139	0.000	0.410	Forward
3	30	0.337	0.324	0.474	0.008	0.000	0.098	Forward
5	30	0.324	0.324	0.335	0.000	0.000	0.000	Forward
9	1	0.324	0.324	0.324	0.000	0.000	0.000	Forward

Table S35. Provider-region and forward-event holdouts. Connected components from the target provider are removed from source fitting/calibration; global targets stop at 2014 and event test years are 2015, 2018 and 2022. All local reconstruction histories are strictly earlier. Regional proxies do not verify hydrological mechanisms.

Region proxy	Train through	Event years	Train G	Cal. G	objective	window	N	G	Retention	Risk	Accepted	Failed	Mean X	Upper X
foen	2014	2015;2018;2022	53	18	quantile	False	210	2	0.686	0.000	144	0	0.016	0.858
foen	2014	2015;2018;2022	53	18	quantile	True	210	2	0.714	0.000	150	0	0.015	0.858
foen	2014	2015;2018;2022	53	18	probability	False	210	2	0.510	0.000	107	0	0.024	0.865
foen	2014	2015;2018;2022	53	18	probability	True	210	2	0.633	0.000	133	0	0.019	0.861
gkd_bayern	2014	2015;2018;2022	48	17	quantile	False	1272	9	0.789	0.029	1004	29	0.033	0.420
gkd_bayern	2014	2015;2018;2022	48	17	quantile	True	1272	9	0.784	0.025	997	25	0.033	0.420
gkd_bayern	2014	2015;2018;2022	48	17	probability	False	1272	9	0.826	0.040	1051	42	0.038	0.428
gkd_bayern	2014	2015;2018;2022	48	17	probability	True	1272	9	0.739	0.011	940	10	0.019	0.390
lubw	2014	2015;2018;2022	51	17	quantile	False	144	7	0.736	0.038	106	4	0.065	0.541
lubw	2014	2015;2018;2022	51	17	quantile	True	144	7	0.667	0.031	96	3	0.057	0.529
lubw	2014	2015;2018;2022	51	17	probability	False	144	7	0.757	0.055	109	6	0.078	0.560
lubw	2014	2015;2018;2022	51	17	probability	True	144	7	0.764	0.055	110	6	0.078	0.559
nve_hydapi	2014	2015;2018;2022	49	17	quantile	False	221	9	0.656	0.007	145	1	0.026	0.405
nve_hydapi	2014	2015;2018;2022	49	17	quantile	True	221	9	0.647	0.007	143	1	0.026	0.406
nve_hydapi	2014	2015;2018;2022	49	17	probability	False	221	9	0.652	0.007	144	1	0.026	0.405
nve_hydapi	2014	2015;2018;2022	49	17	probability	True	221	9	0.661	0.007	146	1	0.025	0.405
usgs	2014	2015;2018;2022	16	6	quantile	False	3628	78	0.658	0.014	2386	33	0.026	0.107
usgs	2014	2015;2018;2022	16	6	quantile	True	3628	78	0.655	0.013	2375	32	0.025	0.104
usgs	2014	2015;2018;2022	16	6	probability	False	3628	78	0.695	0.027	2523	68	0.034	0.119
usgs	2014	2015;2018;2022	16	6	probability	True	3628	78	0.674	0.020	2447	48	0.029	0.112

ence, not proof of an architectural improvement. The developed system supplies its entire empirical H30 law to every point arm, including native tolerance-exceedance probability; extra window features cannot be credited merely for supplying previously omitted direct uncertainty. Group-disjoint development/calibration/testing roles and unavailable original candidates remain explicit. Thirty-two scenarios per case are evaluated using twelve parallel case-chunk workers per role, one torch thread and batch thirty-two each. An optional shared tail scheduler adds thirty-two workers calling exactly the same sampler after byte-agreement checks on reconstructed inputs. Each chunk has a distinct deterministic seed. Repeated-input batching changes random-number ordering but not the conditional sampling law; scheduling cannot change cached chunk outputs. This remains a finite development budget, not a probability-model ceiling.

The Canadian archive is staged with metadata and source-model locks before thermal outcomes. An amended reference definition is written after metadata inspection but before temperature inspection: observed primary days plus independent backup on missing days define a complete recorded year, and both serial orientations are candidates. Sensor agreement and minimum daily coverage are fixed before reconstruction errors. Implementation corrections distinguish moved loggers by reported coordinates and preserve a finite reading when its duplicate is missing; no threshold or historical-support requirement is relaxed. One dependent provincial cohort is retained rather than assigning independent certainty to sensors or repeated references. No originally supported case survives; a separately locked short-history

sensitivity retains all three history-ineligible but reference-compatible pair-years. The later raw-sampling clarification does not exclude any of those references.

Raw sampling audits distinguish absent hourly records from under-covered daily means. All four selected deficient days retain some observations and occur in winter. Illustrative missing-temperature ceilings of 40, 60 and 100 degrees leave the annual H30 upper and observed-window lower bounds equal in these two records. These are conditional support sensitivities, not validated physical or sensor-error bounds; they explain why this sample is weak evidence for peak recovery. True warmer and longer outages with suitable backups remain necessary. Public free backups and air forcing are incorporated or shown as references, never assigned invented query costs.

S11.2 Joint budget design and backup-reference eligibility

The joint sample-budget experiment refits both direct-probability heads, recalibrates thresholds and evaluates paired components in 1,000 repetitions per training size. Original role supports remain disjoint; simulated larger groups are iid draws from their finite empirical role laws, not new observations. Primary gain evidence uses the bounded paired difference in $[-1, 1]$, with one-sided lower bound $\bar{D} - \sqrt{2\log(1/\alpha)/G}$ (Hoeffding, 1963; see the article reference list). Risk tests allocate error separately to both models. With formal calibration, this original paired experiment splits its 5% familywise budget as 0.005 per calibration head (with candidate-family correction), 0.01 per test risk and 0.02 for gain. The empirical selector remains heuristic; independently evaluated test-risk and gain evidence are still required. A normal-bound sensitivity is explicitly non-primary. Monte Carlo intervals are adjusted over the planning grid. Source-like simulations cannot verify independence or represent new hydrological types. The separate future protocol now retains both native-CDF and matched-boosting direct references, requiring joint success against each. Its three risk tests, three formal calibrations and two gain tests share a new 5% error allocation; the original paired source plan does not establish power for that future procedure.

Joint planning for the original priority-point comparison refits learning and calibration 3,000 times across training, calibration and evaluation budgets. Its largest

primary source-like joint success estimate is 3.9%, with no evaluated budget cell supporting the 80% target. Actual role supports are only 59/21/24 components; simulated budgets do not create observed histories or unseen climates. These calculations neither plan the neural-point candidate nor justify recruiting a nominal 59 or 118 groups for it. A separate completed planning experiment now implements the neural-point candidate and both direct references, rather than reusing the original paired plan. The old-only selected air32 probability law and 32 draws remain fixed; both matched boosting heads are refitted on resampled development components in each repetition. All methods share calibration compositions and paired evaluation groups, and score thresholds use training-fixed candidate families. The three formal calibrations and three evaluation-risk tests each receive $\alpha = 0.005$; each of the two gain tests receives 0.01. Their familywise allocation is 0.05. Empirical calibration at its original 0.05 level is reported only as a sensitivity, with the same independent evaluation-risk and paired-gain requirements.

Training budgets 60/120/240, calibration budgets 16/60/120/240/480/960/1920/3840 and evaluation budgets 10/60/120/240/480/960/1920/3840/7680 give 432 cells across the two calibration policies. Three thousand complete paired-head refits produce zero joint successes in each cell's 1,000 repetitions (Table S44). The pointwise 95% Monte Carlo upper bound is 0.37%; the simultaneous 432-cell upper bound is 0.97%. These quantify finite simulation uncertainty, not target-population risk. The actual role supports are only 11/9/7 already opened Swiss components. Independent draws from those finite laws cannot generate new climate, process or regional-event types. No tested budget supports the 80% joint-power target, and no new-population recruitment size is recommended. New Australian reference qualification has not opened target prediction errors and does not turn these simulated budgets into acquired independent units. A newly inspected DFO redundant-logger resource yields no cases within the original three-complete-prior-year support. Three compatible backup-reference pair-years represent only two primary logger-years outside that support. Their four cold-season daily deficits contain 8–19 hourly measurements, rather than demonstrated whole-day failures. Interpolation and the old univariate reference preserve H30 because its hot window remains observed; this is not evidence of difficult peak recovery. These project aggregates and free backup union are separate sensitivities, not warm-season outage confirmation or a costly-

acquisition benefit (Fisheries and Oceans Canada, 2024; see the article reference list).

S11.3 Conditional target-risk laws and complete system attribution

A further experiment on previously opened data separates the conditional H30 law from its delivered numerical point. It retains the same controlled crossed-contiguous Swiss cases as the earlier reconstruction factorial; these masks do not estimate natural outage prevalence. Seven laws retain one priority point, including raw blocks, visible shifts, seasonal slow-plus-short residuals with and without available air, a separate flow tier, and the preserved air32 neural law with and without the same visible shift. Residual laws supply 512 nested draws for sampling diagnostics; all primary law, score and use comparisons use 32 draws. Observations remain fixed and residual histories exclude each target year. This is retrospective supplementation, not a change to the frozen prospective predictor.

CRPS on the common development cases selects the residual model conditioned on season and air temperature. On 907 common old Swiss test cases, its CRPS is 0.210 versus 0.273 degrees Celsius for raw blocks, and nominal-90% coverage is 85.7% versus 71.3% (Tables S45–S47). At the unchanged priority point, its native CDF rule accepts 450 cases with 8 failures, while the corresponding window head accepts 494 with 19 failures. These empirical operating points neither establish the joint five-point advantage against both direct references nor certify risk.

The aggregate improvement does not resolve warm-year errors. Across 357 common warm-year cases at 32 draws, the selected residual law's H30 median bias is -0.379 degrees Celsius and coverage is 77.6%; the neural reference has bias -0.330 and coverage 78.2%. With 512 draws the selected residual law reaches 90.0% aggregate coverage, showing an effect of the finite empirical-band budget. Its warm-year coverage remains 84.6% and bias -0.379 degrees Celsius. These overlapping diagnostic conditions do not identify a physical cause or demonstrate transfer to an unseen climate or event.

The system experiment also changes numerical point while refitting each point's own error and calibrated admission. Four points, two same-air laws and three score types give 24 cells; a separate flow tier adds 18. The 945-candidate denomina-

Table S36. Old-group-only probability development. Both air-conditioned CSDI configurations share data, structured masks, seed, optimizer, maximum thirty epochs and patience five. Selection uses equal-group H30 CRPS on thirty old held-out groups and five masks per group; no external recovery outcome chooses the configuration. These budgets are not a claim of optimized state of the art.

Config.	G	N	Avail.	CRPS C	Epoch cap	Fit minutes	selected
air32	30	150	150	0.261	30.000	51.217	True
air64	30	150	150	0.272	30.000	61.995	False

Table S37. Developed same-air system comparison on all 945 original controlled Swiss candidates. Point availability and admission retain that full pool; MAE compares the same 828 candidates with available Air2stream-3 and conditional distributions. Every point method receives the same selected H30 distribution. Native CDF failure probability is evaluated against that method's point before any window ablation; boosted direct and window scores share capacity and roles. Case risk pools accepted candidates; group risk is the ratio of mean group failure-and-admission to mean group admission. Unavailable numerical points remain unknown. Direct cost classification uses the same H30 law regardless of delivered point. This opened-data comparison is not a new confirmation.

method	Score	Point N	MAE C	Accepted	Failed	Retention	Case risk	Group risk	Formal N
Priority	Native CDF	945	0.346	447	9	0.473	0.020	0.020	0
Priority	Direct head	945	0.346	316	1	0.334	0.003	0.003	0
Priority	Window head	945	0.346	330	3	0.349	0.009	0.009	0
Air2stream-3	Native CDF	828	0.283	449	20	0.475	0.045	0.045	0
Air2stream-3	Direct head	828	0.283	404	11	0.428	0.027	0.027	0
Air2stream-3	Window head	828	0.283	380	7	0.402	0.018	0.018	0
Neural point	Native CDF	945	0.306	474	12	0.502	0.025	0.025	0
Neural point	Direct head	945	0.306	370	6	0.392	0.016	0.016	0
Neural point	Window head	945	0.306	446	16	0.472	0.036	0.036	0

tor is retained for every decision, while common-subset MAE has 828 same-air or 821 flow-supported cases. Improved point error does not by itself validate a CDF tolerance event or admission threshold. All 63 formal rules across the law and system experiments have empty admission. Conditional warm/flow checks and full expected-cost references remain supplementary, with unchanged original H1–H3 and no new independent confirmation.

Table S38. Full-procedure joint planning for the original priority-point direct/window heads under the empirical source laws, 1,000 complete paired refits per training budget. Simulated calibration budgets range from 60 to 3,840 and evaluation budgets from 60 to 7,680 groups. Actual source supports contain 59 fitting, 21 calibration and 24 evaluation components; larger budgets are simulated iid draws, not acquired independent histories. Formal and empirical calibration are both evaluated. Primary success requires risk evidence for both methods and a Hoeffding lower retention-gain bound above five points; the normal approximation is sensitivity only. Each column maximizes over calibration/test budgets. Monte Carlo upper bounds cover the entire planning grid. No evaluated cell supports eighty-percent success. This does not plan a new neural-point candidate.

Train G	Rule	Joint	MC upper	Normal	Population met
60	empirical	0.001	0.012	0.020	0.022
60	formal	0.039	0.068	0.131	0.139
120	empirical	0.001	0.012	0.013	0.017
120	formal	0.027	0.052	0.097	0.105
240	empirical	0.002	0.014	0.005	0.007
240	formal	0.024	0.048	0.071	0.083

Table S39. Redundant-logger resource qualification before reconstruction errors. There are 180 candidate sensor-pair-years, not 180 independent rivers. No case meets the unchanged three-complete-prior-primary-year protocol. Three compatible-reference pair-years correspond to two primary logger-years; they fail only historical support and remain a separately labelled short-history sensitivity. Source quality is project aggregation, not certified daily means.

qualification	pair_years
Annual boundary outside verified primary support	151
Backup does not cover all missing days	11
Fewer than three complete prior primary years	3
Primary year already complete	14
Sensors fail prespecified comparability checks	1

Table S40. Short-history backup-reference sensitivity, retaining all three pair references and two primary years. The four unavailable daily means arise from 8–19 hourly samples per day, occur in winter, and are not demonstrated full-day sensor failures. Their observed hot windows remain intact. Air2stream unavailability is charged as unknown in use, not counted as zero numerical risk. Free backup union completes the recorded reference by definition and establishes no costly-query advantage. All cases form one dependent provincial cohort; they do not validate risk under long warm-season outages.

method	References	Primary years	Available	MAE C	Failures	Available years
Interpolation	3	2	3	0.000	0	2
Air2stream-3	3	2	1	0.000	0	1
Fixed CSDI (8)	3	2	3	0.000	0	2
Developed CSDI	3	2	3	0.000	0	2

Table S41. Developed distribution quality and prefix-sampling sensitivity on available opened cases; unavailable numerical outputs remain explicit in the original-pool system table. Warm-year and low-flow subsets are diagnostic conditions, not additional neural inputs. Low-flow rows require complete forcing coverage; incomplete flow is retained separately. The nominal band is 90 percent, probabilities are raw before Platt calibration, and point shifts compare nested 8/16/32-draw prefixes. Small Monte Carlo shifts cannot establish distributional reliability. Repeated masks are not independent observations.

Role	Draws	Subset	N	G	Bias C	MAE C	CRPS C	Cov. 90	Width C	Brier
Dev.	8	All	1485	11	-0.045	0.264	0.203	0.727	0.730	0.057
Dev.	16	All	1485	11	-0.052	0.261	0.194	0.791	0.785	0.054
Dev.	32	All	1485	11	-0.051	0.258	0.189	0.823	0.744	0.054
Cal.	8	All	1161	9	-0.068	0.218	0.173	0.674	0.475	0.049
Cal.	16	All	1161	9	-0.070	0.212	0.167	0.742	0.533	0.048
Cal.	32	All	1161	9	-0.071	0.212	0.164	0.761	0.501	0.048
Test	8	All	945	7	-0.124	0.347	0.259	0.702	0.841	0.046
Test	16	All	945	7	-0.132	0.340	0.247	0.774	0.957	0.044
Test	32	All	945	7	-0.133	0.329	0.239	0.810	0.931	0.044
Test	32	Warm	378	7	-0.334	0.422	0.304	0.778	0.913	0.109
Test	32	Other years	567	7	0.000	0.267	0.196	0.831	0.944	0.001
Test	32	Low flow	124	7	-0.267	0.390	0.290	0.710	0.927	0.115
Test	32	Other flow	758	7	-0.081	0.303	0.219	0.834	0.945	0.013
Test	32	Incomplete flow	63	7	-0.501	0.519	0.383	0.714	0.772	0.287

Table S42. Developed-system absolute use loss on all 945 original controlled Swiss candidates, at missed-hot cost five and unknown cost 0.05. Native-CDF/direct/window admission heads screen each point separately. Reconstruction, screening and matched-random references are method specific; the calibrated H30-law decision and constant answers are shared across points. Random counts are exact expectations, and unavailable point answers remain unknown. These secondary cost parameters are experimental, and results are opened-data supplements, not new confirmation.

method	Head	Action	N	loss	FN	FP	Unknown
Priority	–	All reconstructed	945	0.370	70.000	0.000	0
Priority	CDF	Magnitude screen	945	0.058	6.000	0.000	498
Shared	–	CDF cost decision	945	0.053	7.000	0.000	301
Shared	–	All unknown	945	0.050	0.000	0.000	945
Shared	–	All hot	945	0.838	0.000	792.000	0
Shared	–	All cool	945	0.810	153.000	0.000	0
Priority	CDF	Matched random	945	0.202	33.111	0.000	498
Priority	Direct	Magnitude screen	945	0.039	1.000	0.000	629
Priority	Direct	Matched random	945	0.157	23.407	0.000	629
Priority	Window	Magnitude screen	945	0.038	1.000	0.000	615
Priority	Window	Matched random	945	0.162	24.444	0.000	615
Air2stream-3	–	All reconstructed	945	0.183	32.000	7.000	117
Air2stream-3	CDF	Magnitude screen	945	0.069	8.000	0.000	496
Air2stream-3	CDF	Matched random	945	0.122	17.353	3.796	496
Air2stream-3	Direct	Magnitude screen	945	0.060	6.000	0.000	541
Air2stream-3	Direct	Matched random	945	0.115	15.614	3.415	541
Air2stream-3	Window	Magnitude screen	945	0.046	3.000	0.000	565
Air2stream-3	Window	Matched random	945	0.111	14.686	3.213	565
Neural point	–	All reconstructed	945	0.271	50.000	6.000	0
Neural point	CDF	Magnitude screen	945	0.067	8.000	0.000	471
Neural point	CDF	Matched random	945	0.161	25.079	3.010	471
Neural point	Direct	Magnitude screen	945	0.030	0.000	0.000	575
Neural point	Direct	Matched random	945	0.136	19.577	2.349	575
Neural point	Window	Magnitude screen	945	0.063	7.000	0.000	499
Neural point	Window	Matched random	945	0.154	23.598	2.832	499

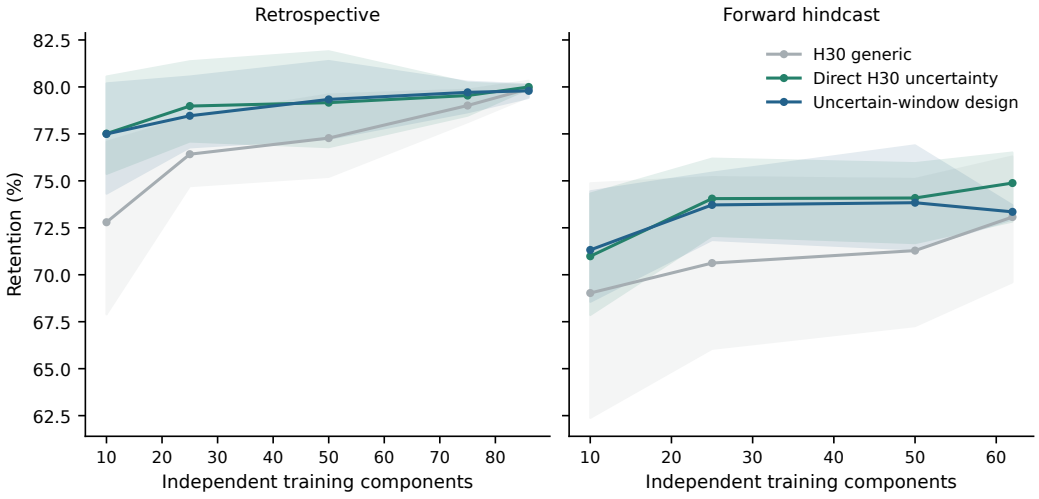


Figure S1. Repeated learning curves. Shading spans the tenth to ninetieth percentiles over training subsets, not new independent rivers or confidence limits. Calibration/evaluation components are fixed. Prospective global fitting/calibration use only years through 2018.

Table S43. Calibration-budget sensitivity for the developed neural point on all 945 controlled Swiss test candidates. All combinations of three, five, seven and nine available calibration groups are included: 84, 126, 36 and one subsets. The old-only probability model, target-development boosted heads and training-fixed threshold families remain unchanged. Native-CDF and boosted scores use identical calibration subsets. Ranges describe subset sensitivity, not confidence intervals, new observations or full-procedure power. Risk summaries omit empty admissions; their counts are explicit. All 741 formal calibrations have zero admission. Loss uses missed-hot cost five and unknown cost 0.05. The full nine-group budget reproduces the original system exactly; no favorable subset is selected.

Cal. G	Score	Sets	Empty	Ret. min	Ret. med	Ret. max	Risk med	Risk max	Loss med
3	Direct head	84	0	0.307	0.419	0.999	0.025	0.246	0.035
3	Window head	84	0	0.429	0.472	0.999	0.036	0.246	0.063
3	Native CDF	84	28	0.000	0.431	0.998	0.025	0.244	0.050
5	Direct head	126	0	0.307	0.392	0.638	0.016	0.103	0.032
5	Window head	126	0	0.429	0.472	0.593	0.036	0.077	0.063
5	Native CDF	126	0	0.431	0.431	0.702	0.010	0.104	0.050
7	Direct head	36	0	0.364	0.392	0.419	0.016	0.025	0.031
7	Window head	36	0	0.472	0.472	0.525	0.036	0.050	0.063
7	Native CDF	36	0	0.431	0.431	0.600	0.010	0.062	0.050
9	Direct head	1	0	0.392	0.392	0.392	0.016	0.016	0.030
9	Window head	1	0	0.472	0.472	0.472	0.036	0.036	0.063
9	Native CDF	1	0	0.502	0.502	0.502	0.025	0.025	0.067

Table S44. Three-method, two-reference joint planning for the developed neural H30 point. Each training-size/policy row summarizes 72 calibration/evaluation budget cells, with 1,000 complete score-head refits, recalibrations and evaluations per cell. The reconstruction law remains frozen as specified for F2. Window boosting must exceed both native-CDF and matched direct boosting by five retention points while all three satisfy risk evidence with positive admission. The target fraction uses the finite role-specific simulation population, whereas hits additionally require evaluation bounds. MC upper is the largest pointwise 95-percent binomial Monte Carlo upper bound; the separate 432-cell-adjusted upper bound is 0.0097 when all hits are zero. Simulated group counts are not newly observed groups. The empirical calibration is a sensitivity only. Actual old development/calibration/test supports are 11/9/7 components; no new-target recruitment size or independent confirmation follows.

Train G	Calibration	Cells	Max hits/cell	Max target fraction	MC upper
60	Empirical sensitivity	72	0/1000	0.009	0.004
60	Formal	72	0/1000	0.041	0.004
120	Empirical sensitivity	72	0/1000	0.028	0.004
120	Formal	72	0/1000	0.042	0.004
240	Empirical sensitivity	72	0/1000	0.021	0.004
240	Formal	72	0/1000	0.036	0.004

Table S45. Conditional H30-law quality on identical controlled Swiss cases at 32 draws. All laws evaluate the same original priority point's tolerance event; median error describes a possible alternative numerical point, not a change to that fixed-point comparison. The common subset excludes unsupported laws only by declared availability; full-pool admission and costs retain all 945 candidates. Temperature errors and widths are in degrees Celsius; coverages are fractions. The flow law adds information and is supplementary. Laws and neural configuration were developed on already opened data; these controlled gaps do not estimate natural outage prevalence or provide new independent confirmation.

Law	N	Median MAE	Bias	CRPS	Cover.	Width	Failure BS	Hot BS
Raw blocks	907	0.361	-0.200	0.273	0.713	0.684	0.185	0.063
Visible shift	907	0.340	-0.224	0.254	0.731	0.689	0.167	0.059
Seasonal slow	907	0.294	-0.086	0.210	0.838	0.910	0.124	0.049
Seasonal + air	907	0.291	-0.083	0.210	0.857	0.928	0.123	0.050
Seasonal + air + flow	907	0.273	-0.084	0.195	0.860	0.914	0.118	0.049
Neural air32	907	0.329	-0.131	0.238	0.811	0.943	0.137	0.040
Neural + visible shift	907	0.329	-0.141	0.238	0.806	0.940	0.133	0.036

Table S46. Conditional diagnostics on the common law-support subset at 32 draws. The existing warm-anomaly flag requires recorded H30 at least one degree Celsius above the permitted training climatology's H30; it differs from the local-q90 label used for costs. Low flow requires complete gap-flow observations and at least half of missing-water days at or below the permitted training month's discharge twentieth percentile. Incomplete flow remains unknown. Conditions overlap and masks do not create independent events or physical attribution. The neural reference and residual candidate were selected using old development CRPS only.

Law	Condition	N	Bias	CRPS	Cover.	Failure BS
Seasonal + air	Warm year	357	-0.379	0.269	0.776	0.151
Seasonal + air	Low-flow gap	124	-0.341	0.261	0.774	0.172
Seasonal + air	Incomplete flow	42	-0.702	0.457	0.738	0.214
Seasonal + air + flow	Warm year	357	-0.366	0.260	0.782	0.149
Seasonal + air + flow	Low-flow gap	124	-0.304	0.247	0.766	0.160
Seasonal + air + flow	Incomplete flow	42	-0.702	0.457	0.738	0.214
Neural air32	Warm year	357	-0.330	0.302	0.782	0.132
Neural air32	Low-flow gap	124	-0.267	0.290	0.710	0.178
Neural air32	Incomplete flow	42	-0.554	0.407	0.714	0.131

Table S47. Nested 32/128/512-draw sensitivity for residual laws on common opened Swiss support. Admission comparisons use only the first 32 draws for every law. The same fixed laws and cases isolate the finite sampling budget: aggregate empirical-band coverage can approach its nominal level as draws increase, while the warm-year columns retain conditional errors. Warm cover and bias use the same 357 common warm-year cases. Larger scenario sets do not supply new river events or verify transfer. Neural laws have only their preserved 32 draws and are not represented as 128- or 512-draw experiments.

Law	Draws	N	CRPS	Cover.	Width	Failure BS	Warm cover.	Warm bias
Raw blocks	32	907	0.273	0.713	0.684	0.185	0.605	-0.487
Raw blocks	128	907	0.272	0.734	0.731	0.181	0.625	-0.492
Raw blocks	512	907	0.270	0.749	0.747	0.180	0.630	-0.490
Visible shift	32	907	0.254	0.731	0.689	0.167	0.633	-0.457
Visible shift	128	907	0.253	0.749	0.736	0.166	0.636	-0.462
Visible shift	512	907	0.251	0.771	0.752	0.165	0.661	-0.460
Seasonal slow	32	907	0.210	0.838	0.910	0.124	0.759	-0.384
Seasonal slow	128	907	0.205	0.879	0.976	0.122	0.810	-0.382
Seasonal slow	512	907	0.205	0.894	0.996	0.121	0.829	-0.381
Seasonal + air	32	907	0.210	0.857	0.928	0.123	0.776	-0.379
Seasonal + air	128	907	0.203	0.888	0.992	0.121	0.818	-0.379
Seasonal + air	512	907	0.203	0.900	1.012	0.121	0.846	-0.379
Seasonal + air + flow	32	907	0.195	0.860	0.914	0.118	0.782	-0.366
Seasonal + air + flow	128	907	0.189	0.896	0.976	0.115	0.826	-0.366
Seasonal + air + flow	512	907	0.189	0.910	0.995	0.116	0.852	-0.367

Table S48. Fixed-priority-point admission for seven H30 laws and three matched score types. A/F is accepted/failed, with failure defined by absolute priority-point error above 0.5 degrees Celsius. All 945 original candidates remain, including unsupported laws as unknown. Boosted heads share the declared common fitting/calibration support and generic feature budget; the window head adds only the corresponding law's window summaries. Values use the empirical calibration rule; all 21 multiplicity-adjusted formal rules give no admission, whose risk is undefined. Sample operating risk is not a population guarantee.

Law	CDF available	Native A/F	Direct A/F	Window A/F
Raw blocks	945	0/0	376/1	440/13
Visible shift	945	0/0	395/5	398/5
Seasonal slow	907	461/15	495/25	498/22
Seasonal + air	907	450/8	553/40	494/19
Seasonal + air + flow	907	448/9	521/29	465/11
Neural air32	945	447/9	363/4	437/22
Neural + visible shift	945	446/10	372/3	388/11

Table S49. Complete same-air point/law/score system comparison. Each row contains three own-error heads, hence 24 cells across four points and two development-selected laws. Matched MAE uses the identical 828 supported test cases. Every admission and loss uses all 945 original candidates, charging unavailable outputs as unknown. Losses use missed-hot cost five and unknown cost 0.05; always unknown therefore has loss 0.05. Window loss uses numerical screening followed by classification; cost-rule loss directly compares cool, hot and unknown expected costs. All references, random-rejection expectations and cost grids are retained in the source CSV. Empirical operating points do not certify risk, and all formal admissions remain empty.

Point	CDF law	Matched MAE	Native A/F	Direct A/F	Window A/F	Window loss	Cost-rule loss
Priority	Neural air32	0.346	447/9	369/14	324/8	0.033	0.053
Priority	Seasonal + air	0.346	450/8	475/23	410/4	0.060	0.048
Air2stream-3	Neural air32	0.283	449/20	369/4	363/4	0.041	0.053
Air2stream-3	Seasonal + air	0.283	462/21	435/11	387/6	0.061	0.048
Neural median	Neural air32	0.306	474/12	380/6	430/15	0.075	0.053
Neural median	Seasonal + air	0.306	568/40	590/48	589/45	0.072	0.048
State median	Neural air32	0.240	454/14	230/1	383/6	0.067	0.053
State median	Seasonal + air	0.240	567/40	508/33	438/15	0.101	0.048

Table S50. Flow-tier system comparison, separately from the same-air tier. Each row contains native-CDF, direct and window heads, giving 18 cells across three numerical points and two laws. Matched MAE uses 821 common supported cases; admission and utility retain all 945 candidates. The flow-conditioned law and Air2stream-8 add observed discharge information and do not identify an architectural effect. A/F denotes accepted/failed, with the same 0.5-degree event. The illustrated loss uses missed-hot cost five and unknown cost 0.05. All formal rules have empty admission.

Point	CDF law	Matched MAE	Native A/F	Direct A/F	Window A/F	Window loss	Cost-rule loss
Priority	Seasonal + air	0.342	450/8	421/7	416/8	0.054	0.048
Priority	Seasonal + air + flow	0.342	448/9	379/2	393/3	0.056	0.032
Air2stream-8	Seasonal + air	0.203	462/17	403/8	400/8	0.078	0.048
Air2stream-8	Seasonal + air + flow	0.203	493/23	474/17	411/11	0.082	0.032
Flow-state median	Seasonal + air	0.226	571/36	433/12	375/5	0.051	0.048
Flow-state median	Seasonal + air + flow	0.226	608/46	444/11	426/12	0.064	0.032

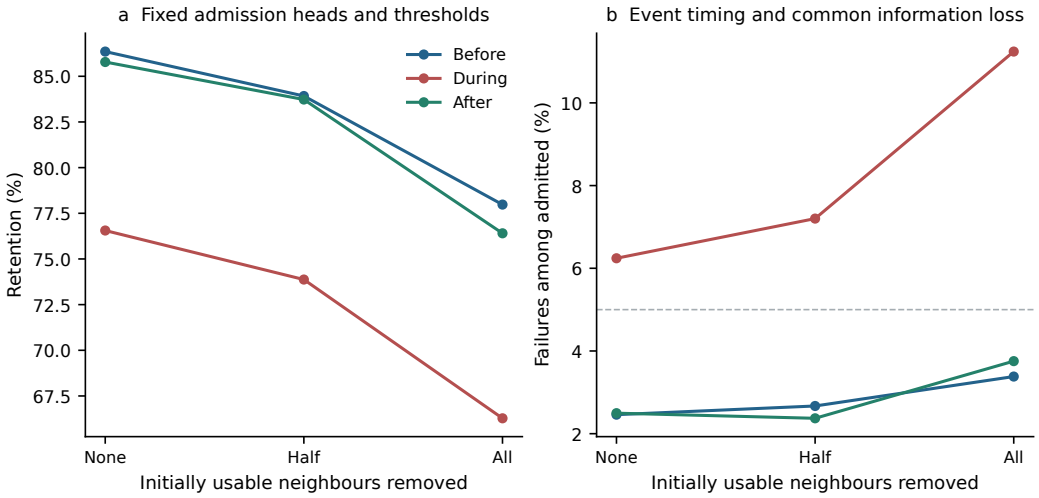


Figure S2. Nondegenerate opened-source event/cofailure stress with unchanged admission heads and thresholds. Values average three split sensitivities, not independent experiments. Timing is relative to the same thermal episode. The independent European station bundles lack donor variation, so this plot is not independent target confirmation.

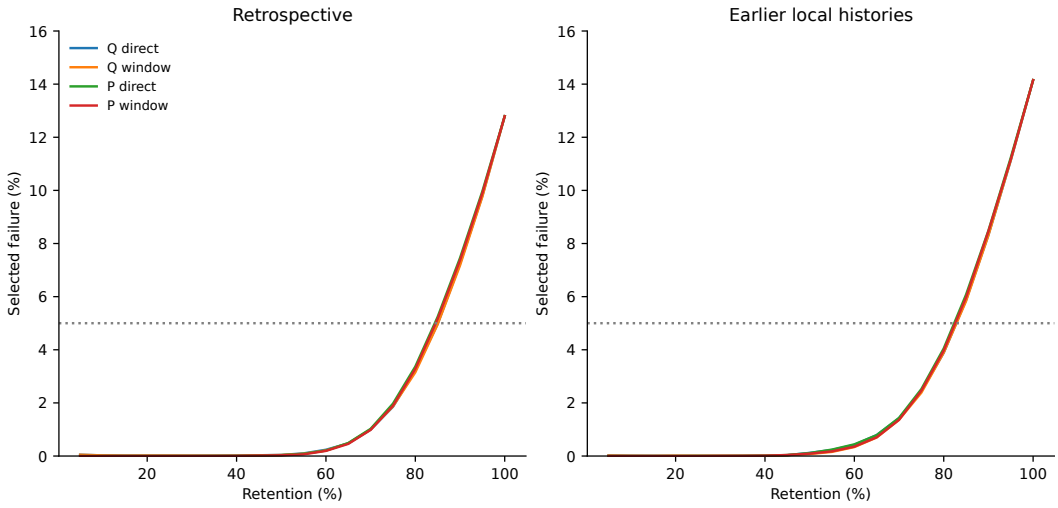


Figure S3. Full risk–retention relations for the four matched learning cells, averaged over the three group-split seeds. These are opened-data sensitivity results; curves neither validate the nominal risk cap nor imply that seeds are independent replications.

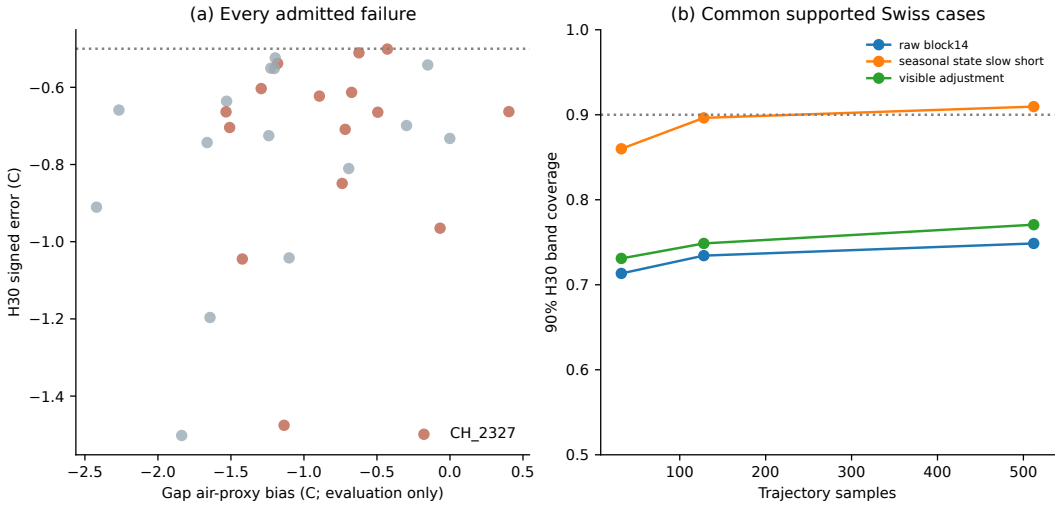


Figure S4. Supplementary failure and scenario checks. (a) Gap-period air-proxy bias versus H30 error for all original admitted failures with available proxy values, highlighting CH_2327. Hidden-day bias is an evaluation diagnostic. (b) H30 scenario coverage under nested sample counts on the common supported Swiss test subset. These model-development results do not establish general conditional coverage or a new independent improvement.